IT3021
Data Warehousing & Business Intelligence

Assignment 2

Submitted By: Rajapaksha R P H A

IT23541184

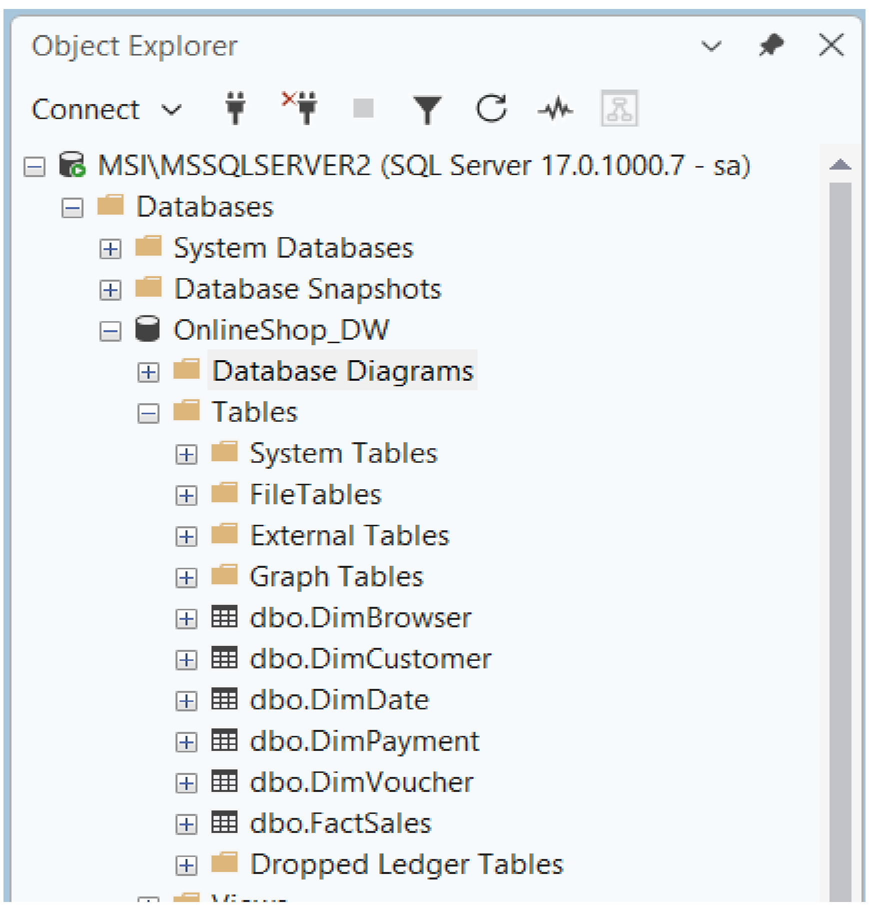
1. Data Source for Assignment 2

The data source used for Assignment 2 is the data warehouse implemented and populated in Assignment 1. The warehouse was created in SQL Server under the database named OnlineShop_DW. This data warehouse is designed using a dimensional model to support analytical processing, reporting, and business intelligence tasks.

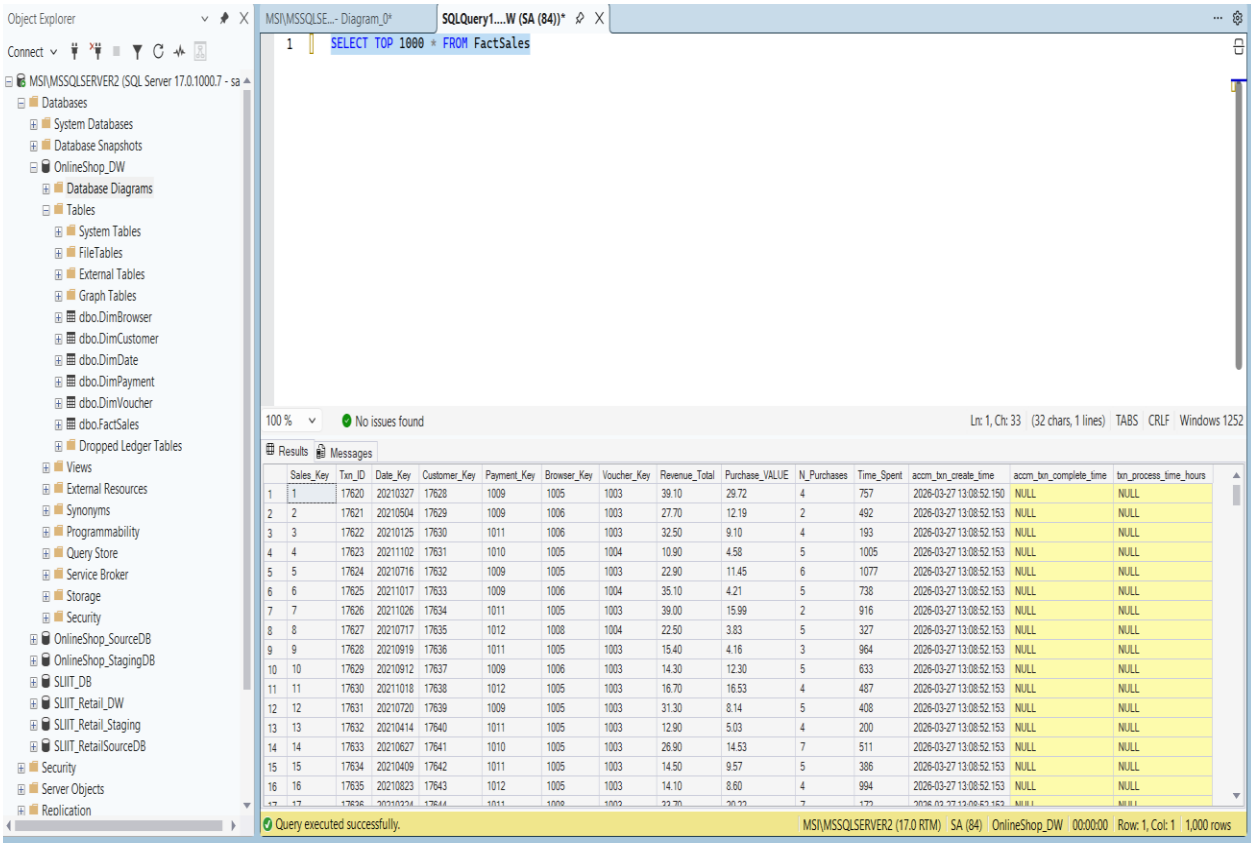
The main fact table in the warehouse is dbo.FactSales. This table stores the quantitative measures required for analysis, such as sales-related values and activity measures. The grain of the fact table is defined as one row per online shop sales record.

The warehouse also contains the following dimension tables: dbo.DimDate, dbo.DimCustomer, dbo.DimPayment, dbo.DimBrowser, and dbo.DimVoucher. These dimensions provide the descriptive context needed for analysis. The dbo.DimDate table supports time-based analysis, dbo.DimCustomer stores customer-related attributes, dbo.DimPayment stores payment-related details, dbo.DimBrowser stores browser-related information, and dbo.DimVoucher stores voucher-related information.

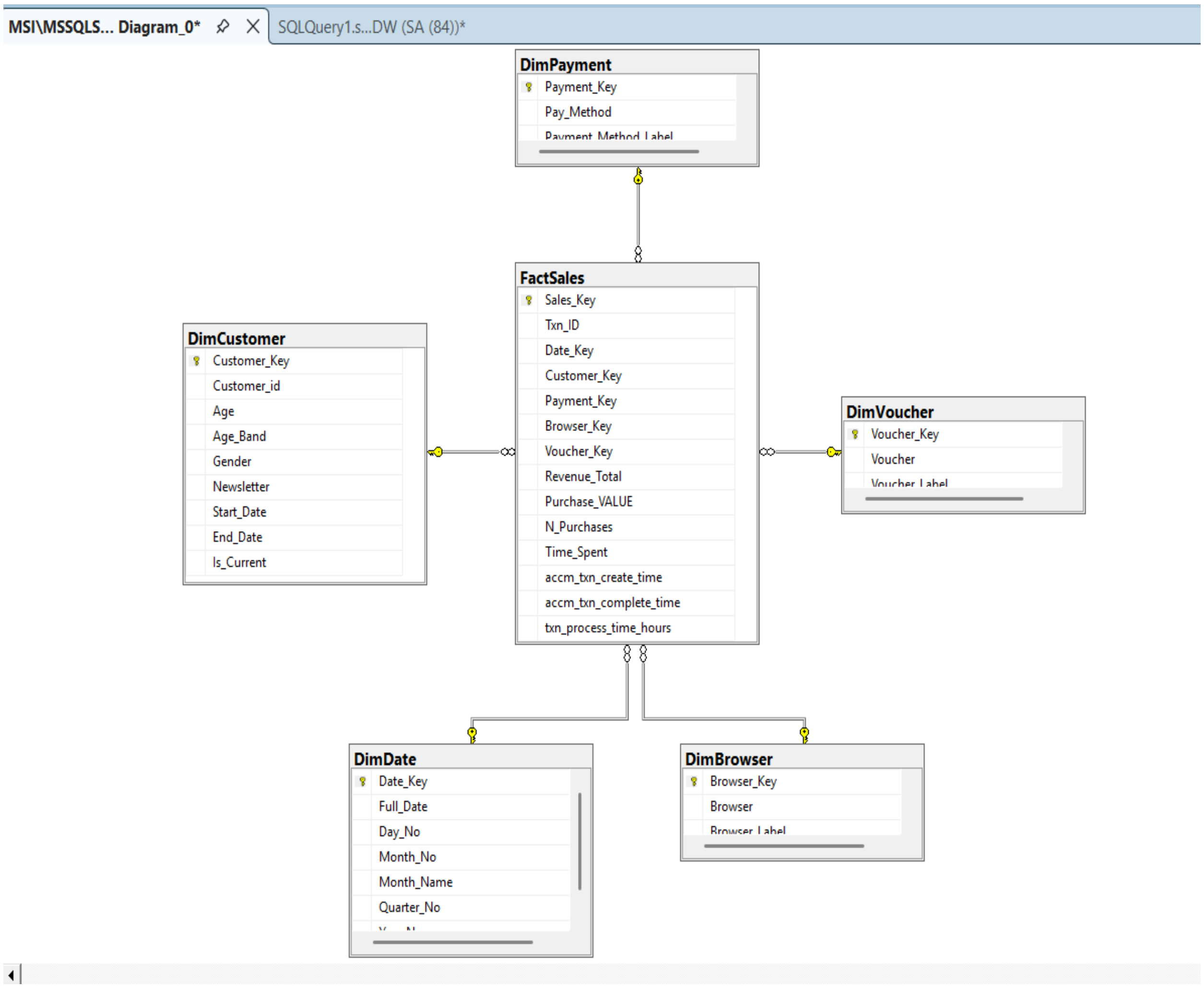
This data warehouse serves as the data source for Assignment 2 tasks including SSAS cube creation, OLAP analysis in Excel, and Power BI report development.



OnlineShop_DW database used as the data source for Assignment 2, containing the fact table FactSales and the related dimension tables DimDate, DimCustomer, DimPayment, DimBrowser, and DimVoucher.



This screenshot shows the FactSales table in the OnlineShop_DW data warehouse after executing a query in SQL Server Management Studio. It confirms that the fact table has been loaded successfully and the sales records are available for analysis.



This diagram shows the star schema design of the OnlineShop_DW data warehouse. It illustrates how the central FactSales table is connected to the dimension tables such as Customer, Date, Payment, Browser, and Voucher for analytical reporting.

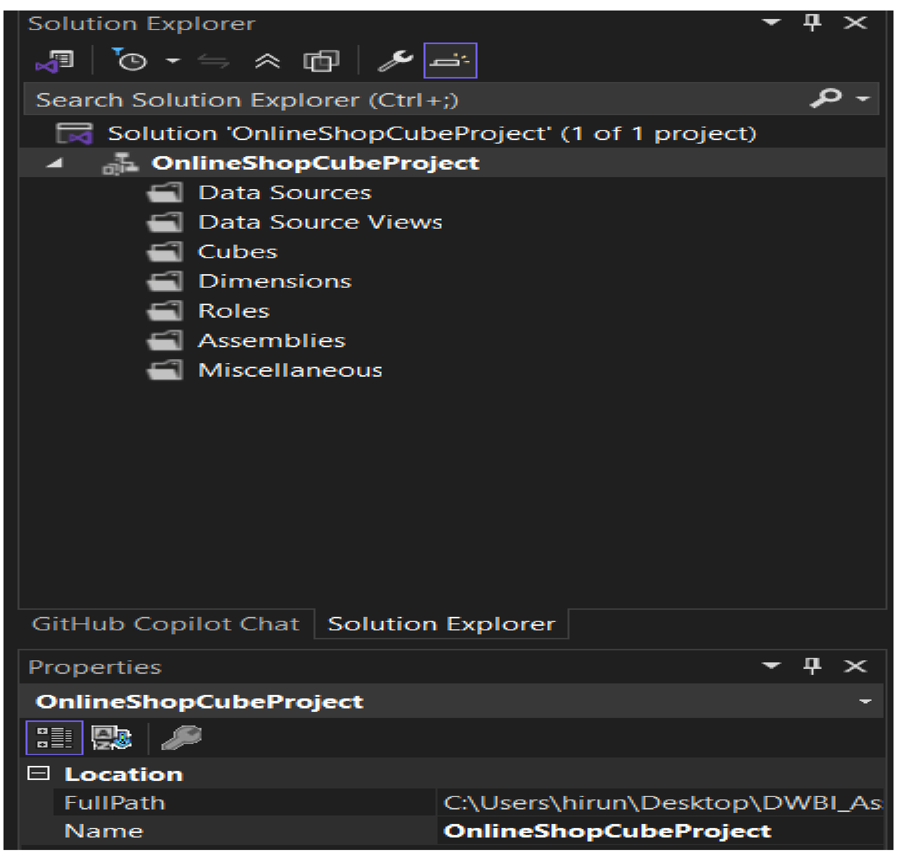
2. SSAS Cube implementation

An SSAS multidimensional project was created in Visual Studio using the data warehouse OnlineShop_DW as the data source. A data source and a data source view were created to connect the project to the warehouse tables FactSales, DimDate, DimCustomer, DimPayment, DimBrowser, and DimVoucher. A cube named OnlineShopCube was then created using the Cube Wizard. The fact table FactSales was selected as the measure group table, and the main numeric columns were selected as measures.

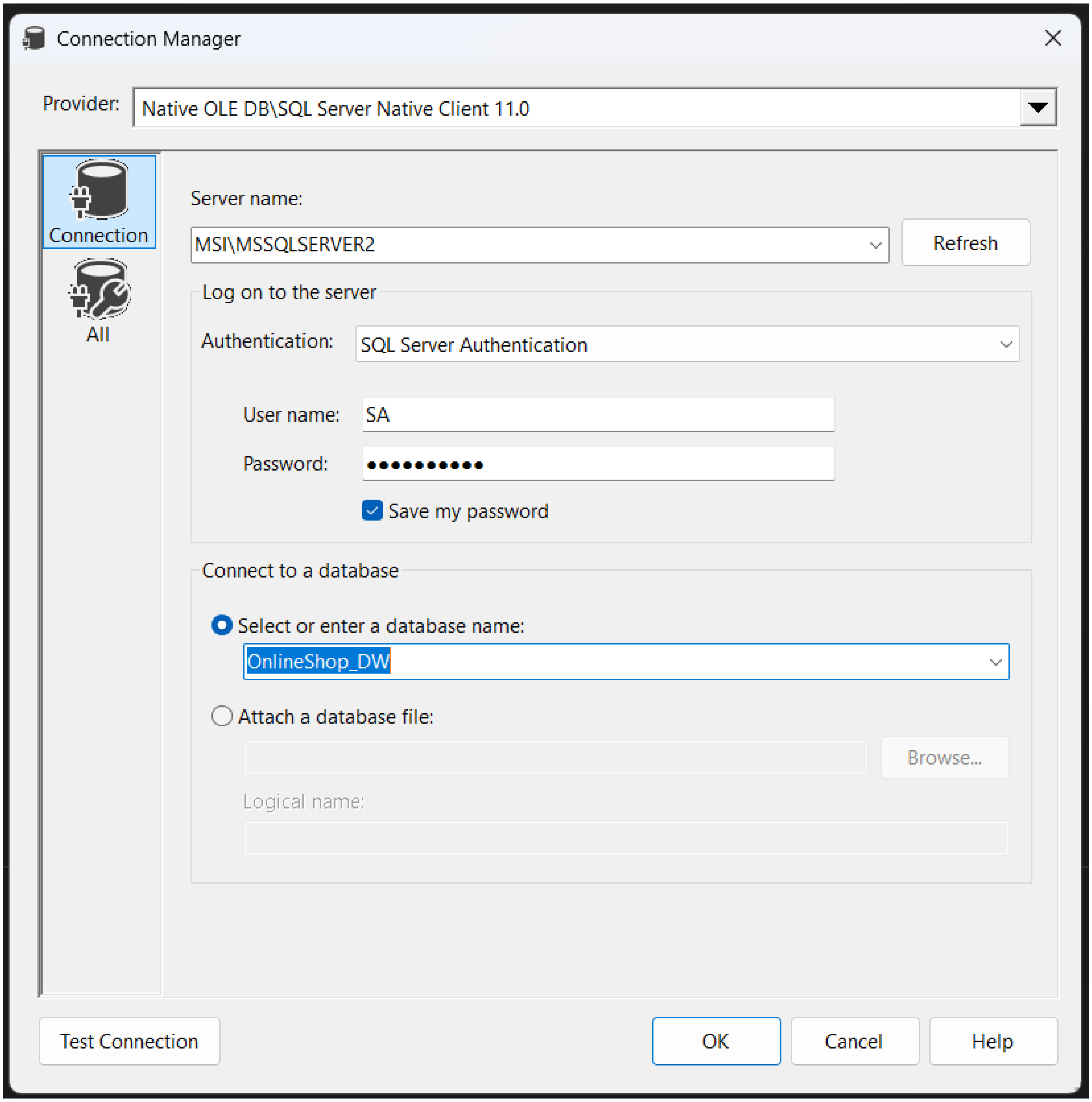
The related dimension tables were added to the cube to support multidimensional analysis. To satisfy the hierarchy requirement, a user-defined hierarchy named Calendar was created in the DimDate dimension using levels such as Year, Quarter, Month, and Date. After reviewing the cube structure, the project was deployed to the local SQL Server Analysis Services instance and processed successfully. The browser tab was then used to confirm that the cube could be queried correctly.

2.1. Create the SSAS project

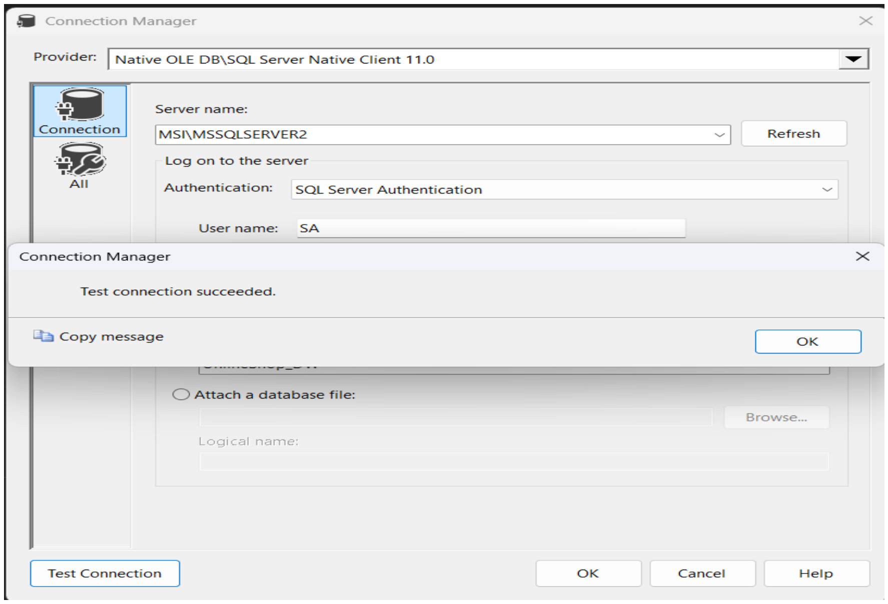
This screenshot shows the Solution Explorer after creating the SSAS project in Visual Studio. It confirms that the project structure was successfully generated with the main folders needed to build the data source, data source view, cubes, dimensions, and other analysis objects.



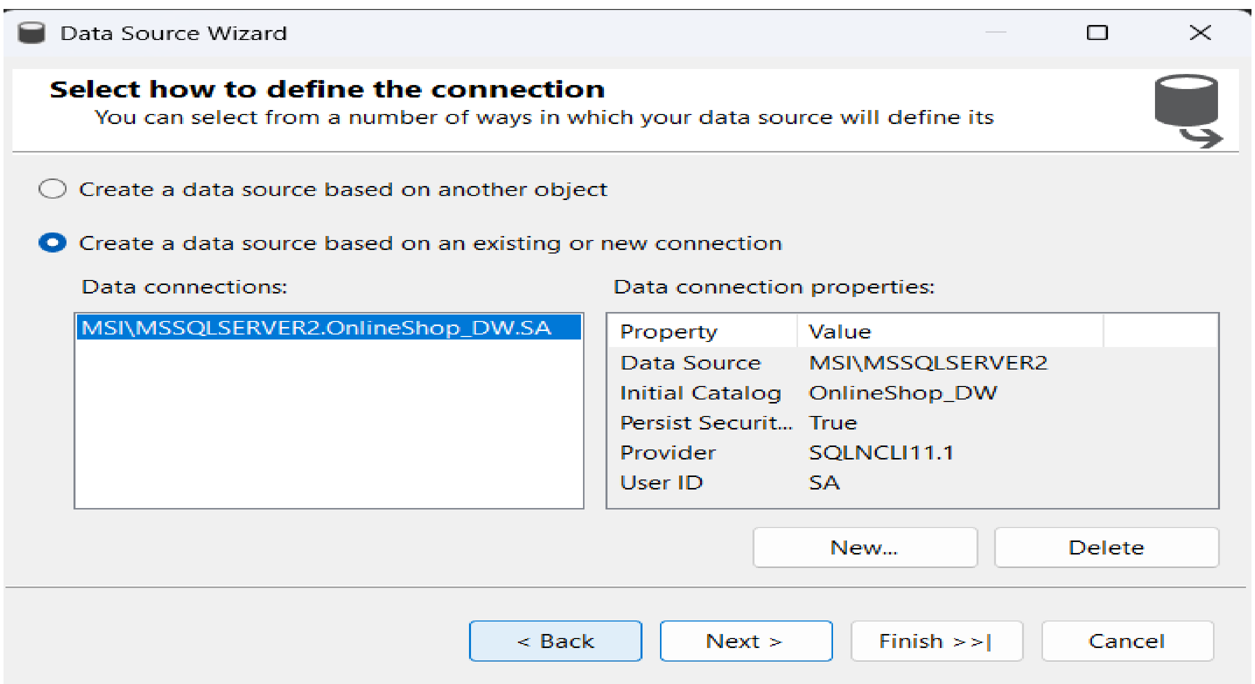
2.2.Create the Data Source



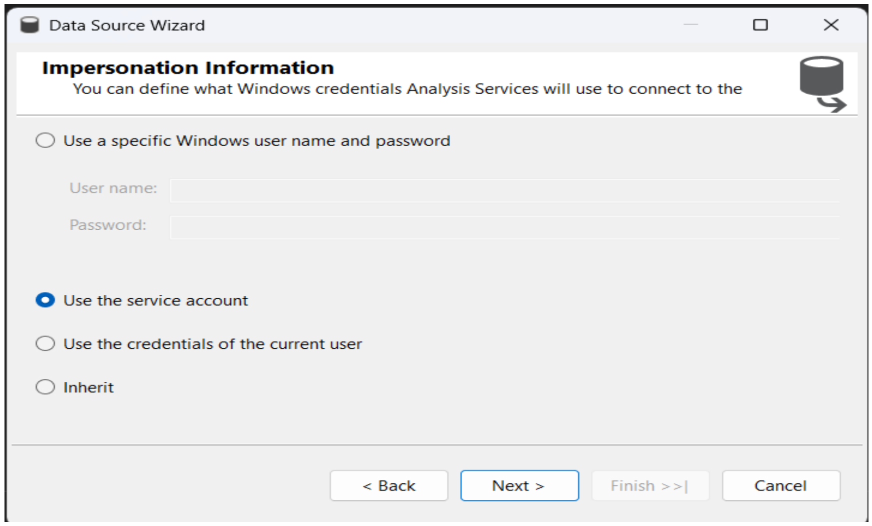
- This screenshot shows the database connection setup for the SSAS project. It confirms that the project was connected successfully to the OnlineShop_DW data warehouse using SQL Server Authentication.



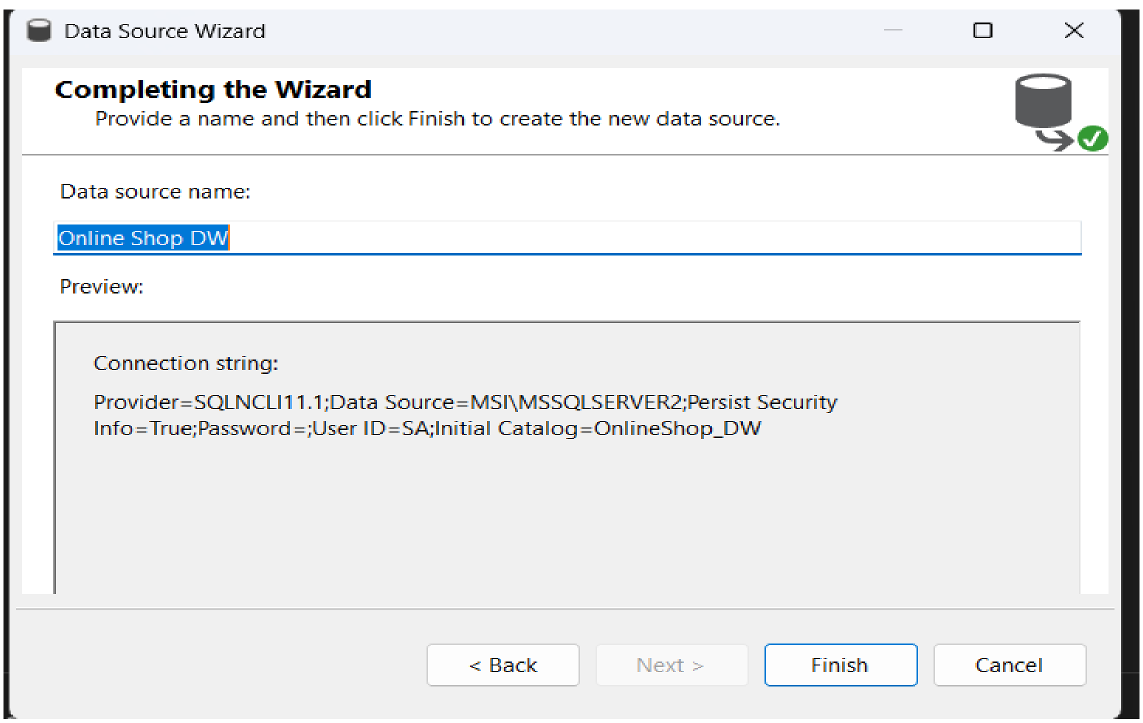
- This screenshot confirms that the connection to the OnlineShop_DW database was tested successfully. It shows that the SSAS project can communicate properly with the SQL Server data warehouse.



- This screen shows the Data Source Wizard step where the connection for the SSAS project is selected. Here, the existing connection to the OnlineShop_DW data warehouse on MSI\MSSQLSERVER2 has been chosen, which will be used as the source for building the cube and related objects.

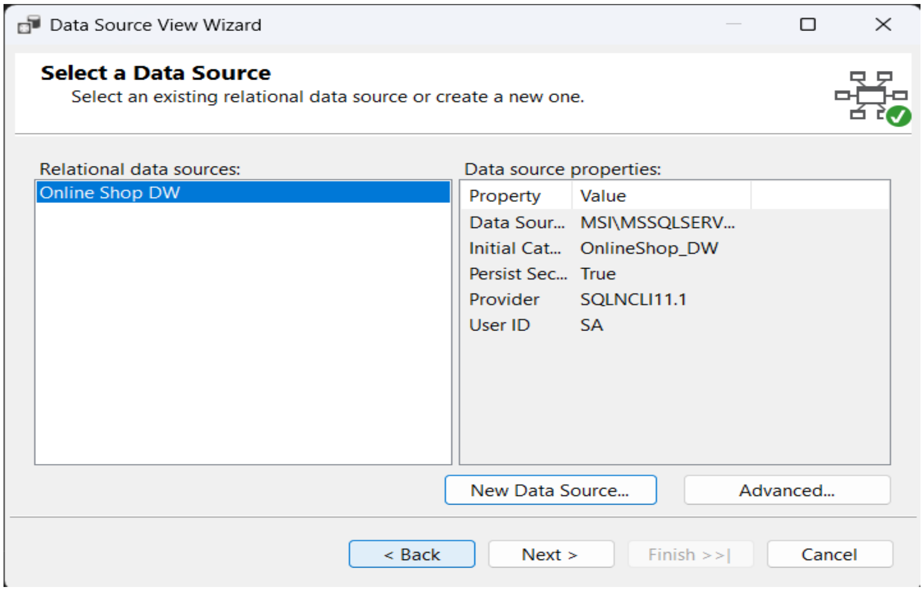


- This screen shows the Impersonation Information step in the Data Source Wizard. Here, Use the service account is selected, meaning Analysis Services will use its service account credentials to access the data source during processing.

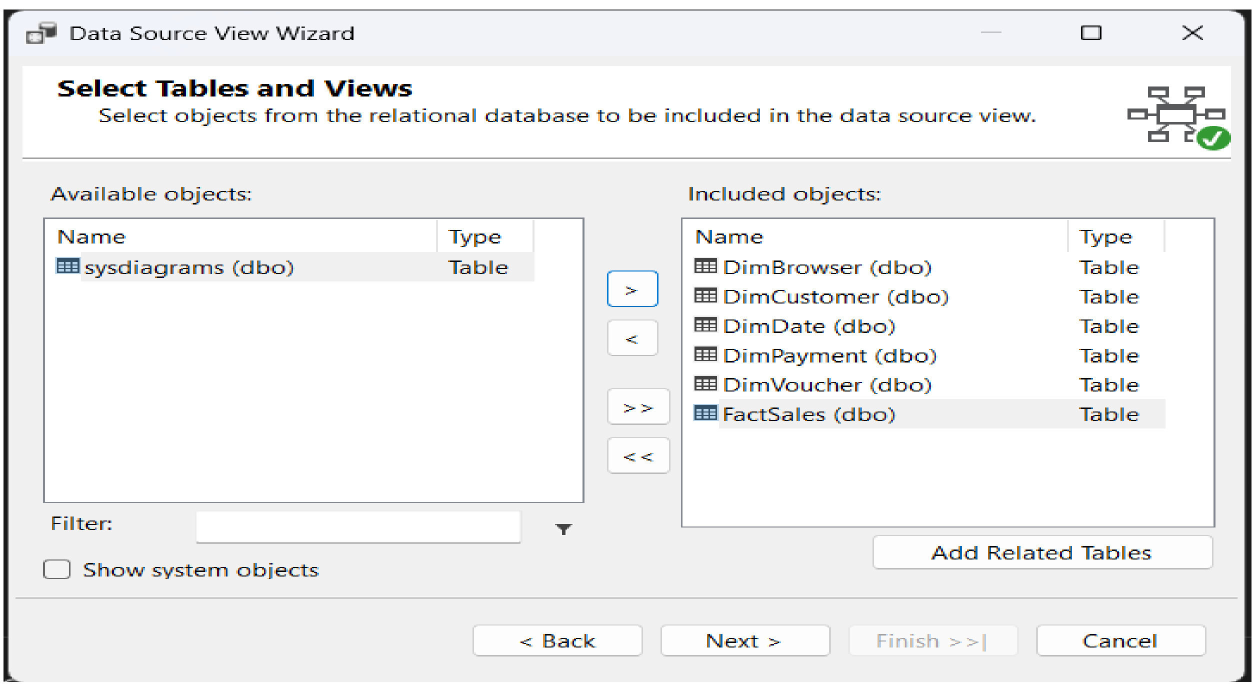


- This screen shows the final step of the Data Source Wizard, where the data source is given the name Online Shop DW. It confirms the selected connection details before finishing and creating the data source for the SSAS project.

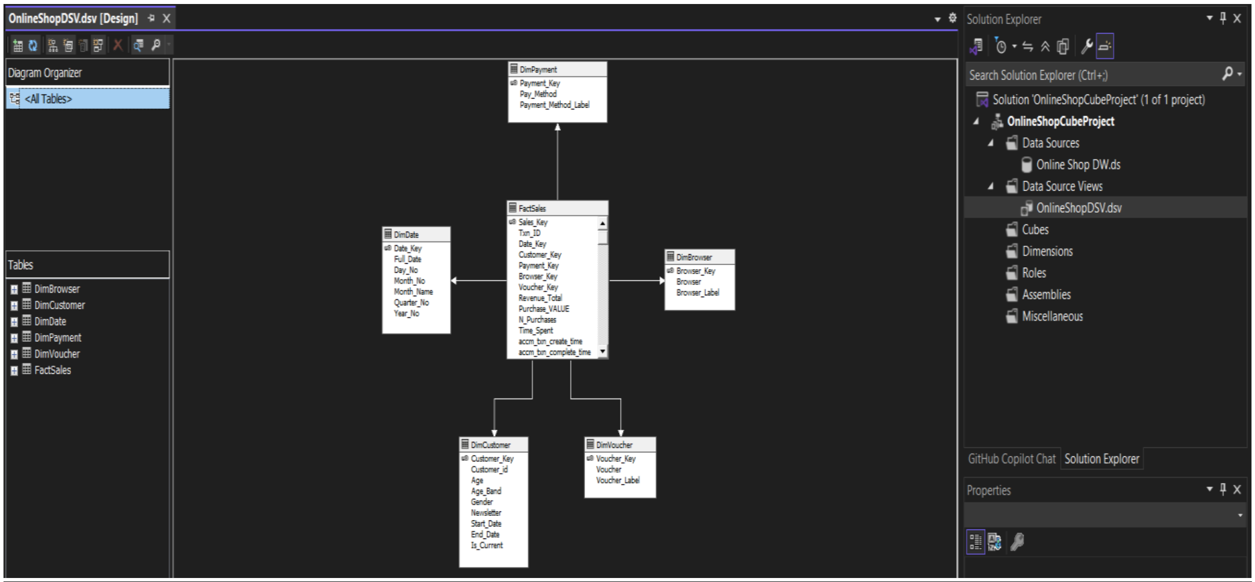
2.3.Create the Data Source View (DSV)



- This screen shows the **Data Source View Wizard** step where the relational data source is selected. Here, the **Online Shop DW** data source is chosen to create the data source view that will be used for the SSAS project.

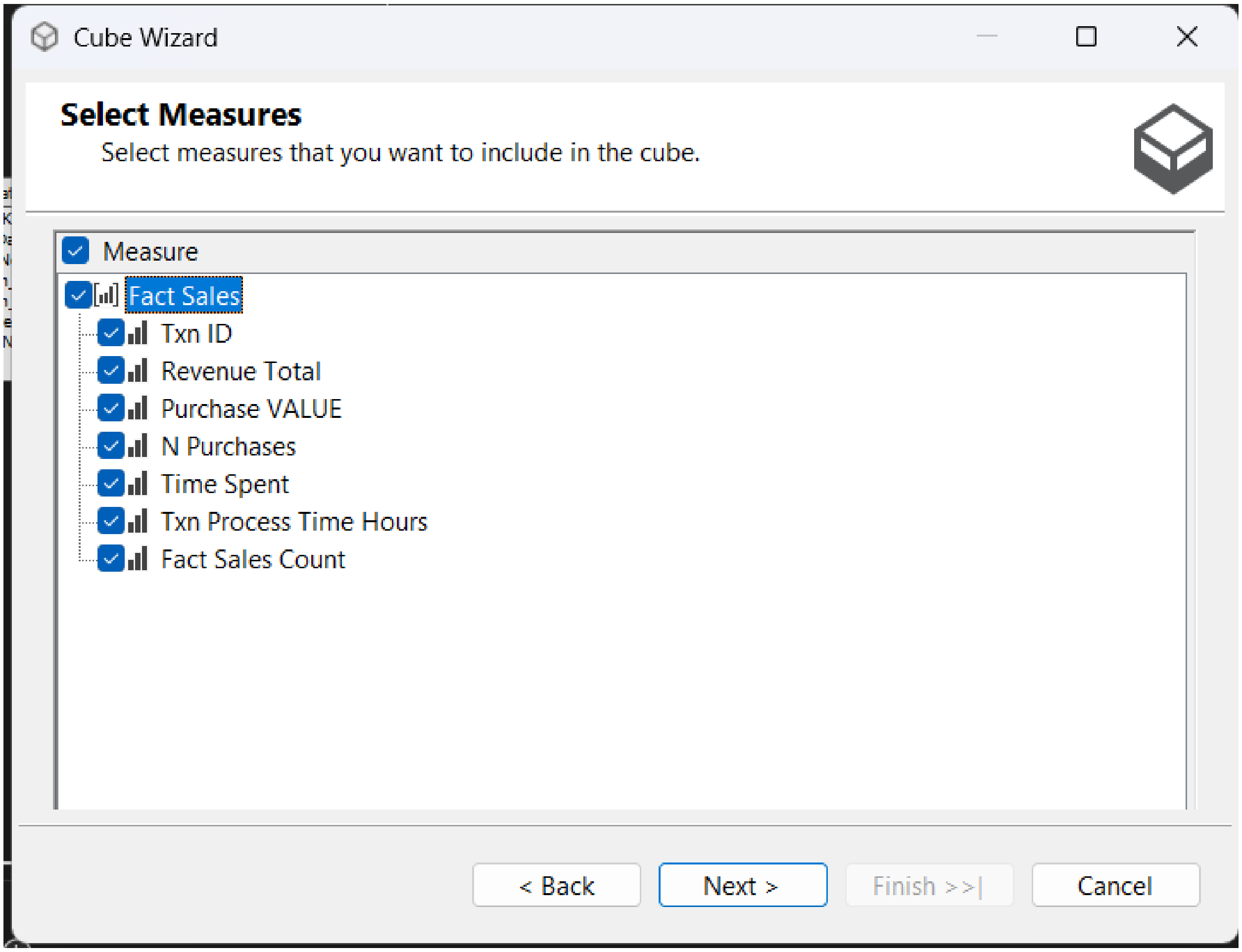


- This screen shows the Select Tables and Views step in the Data Source View Wizard. Here, the required dimension tables and the FactSales table have been added to the data source view for building the SSAS cube structure.

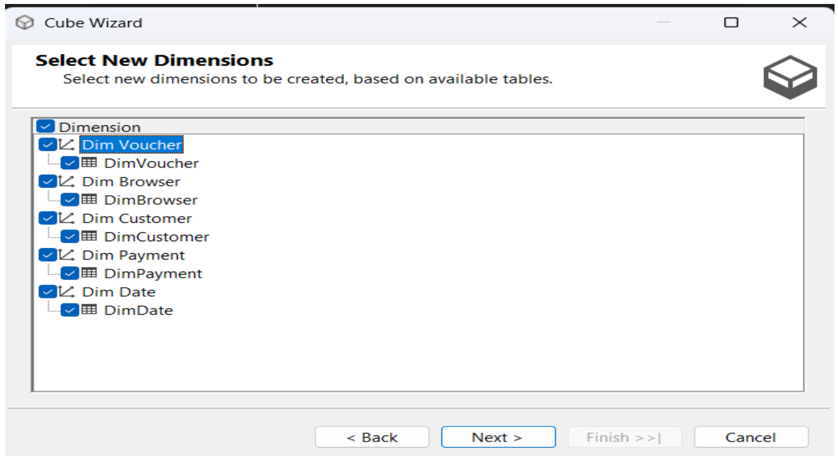


- This screen shows the completed **Data Source View (DSV)** in Visual Studio for the SSAS project. It displays the **FactSales** table linked with its related dimension tables, forming the star schema that will be used to design the cube.

2.4.Create the cube

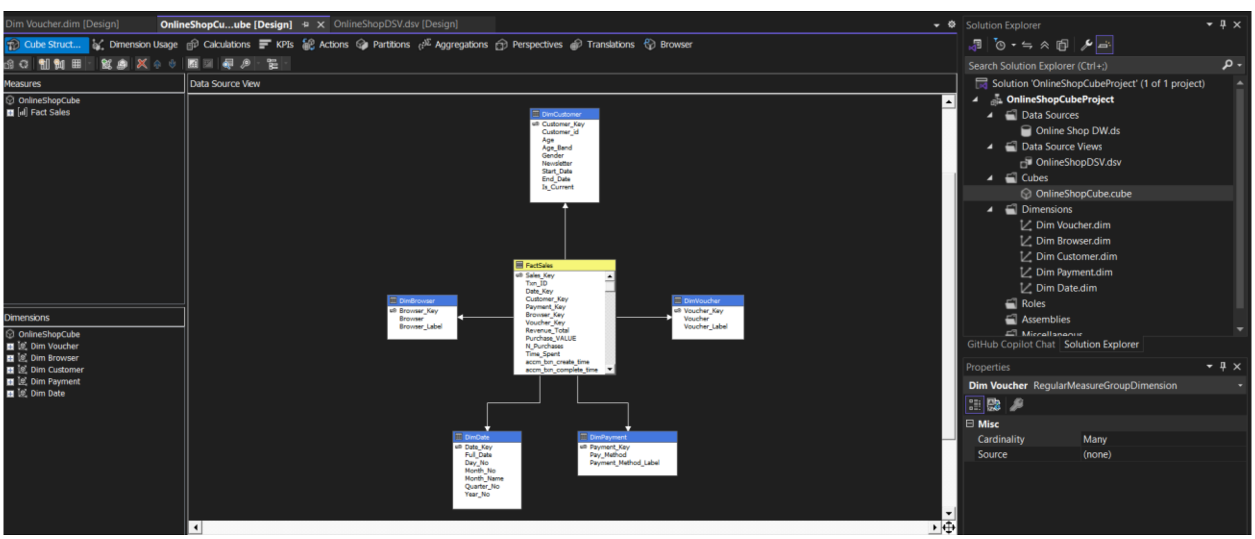


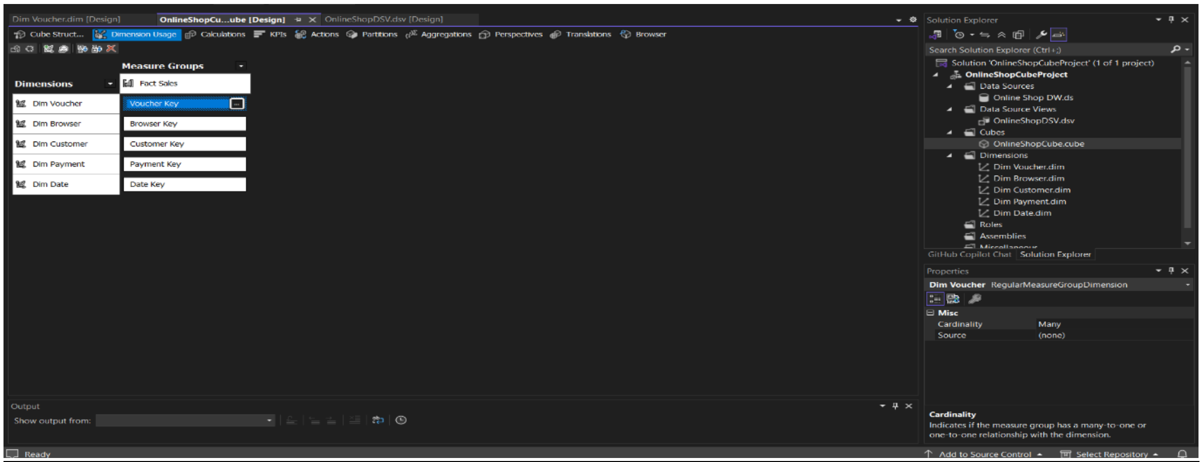
- This screen shows the **Select Measures** step in the Cube Wizard. Here, the required numeric fields from the **Fact Sales** table are selected as measures which will be used in the cube for analysis and reporting.



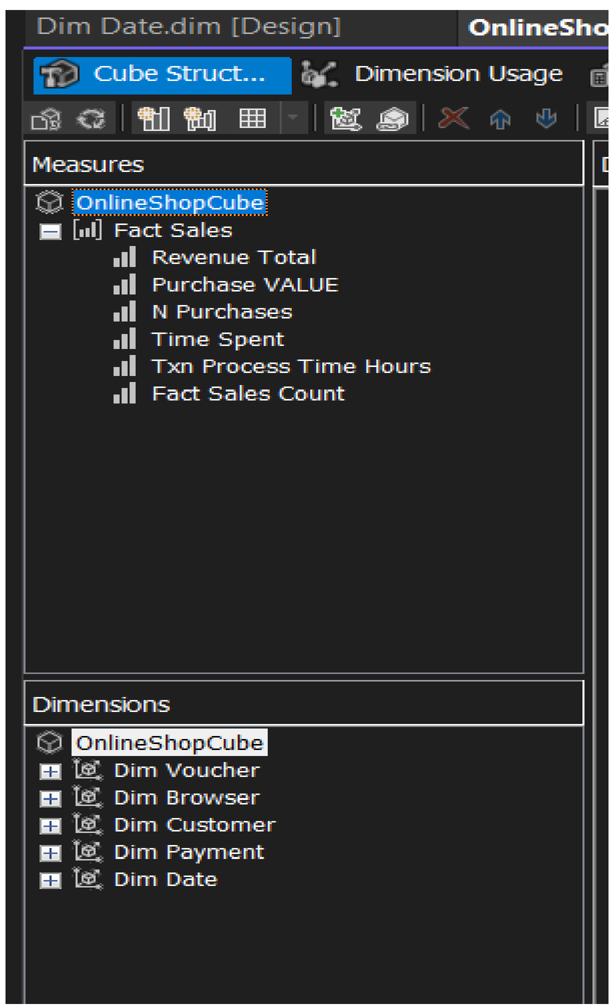
- This screen shows the **Select New Dimensions** step in the Cube Wizard. Here, the dimension tables such as **Dim Voucher, Dim Browser, Dim Customer, Dim Payment,** and **Dim Date** are selected to be created and linked with the cube for multidimensional analysis.

2.5.Review the cube structure



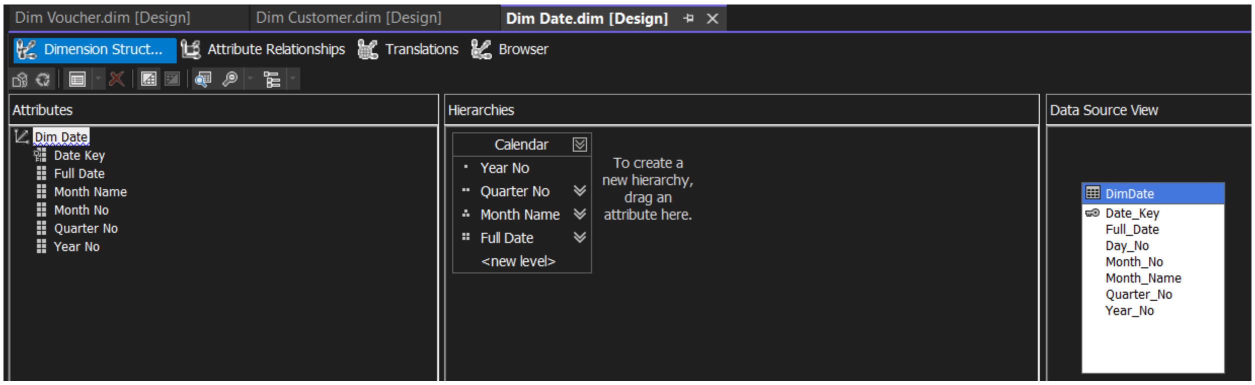


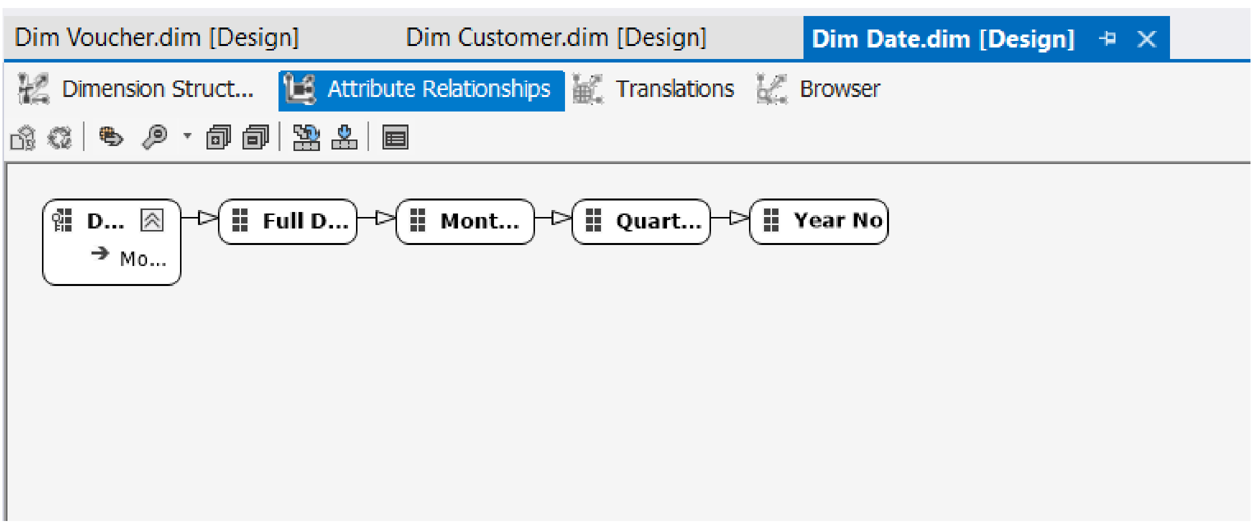
- These images show the **Cube Structure** and **Dimension Usage** views of the SSAS cube design. The first image presents the star schema inside the cube, where the **FactSales** table is connected to all related dimension tables. The second image confirms that each dimension is properly linked to the **Fact Sales** measure group through its corresponding key, defining the relationships used for cube analysis.



2.6. Creating two hierarchies

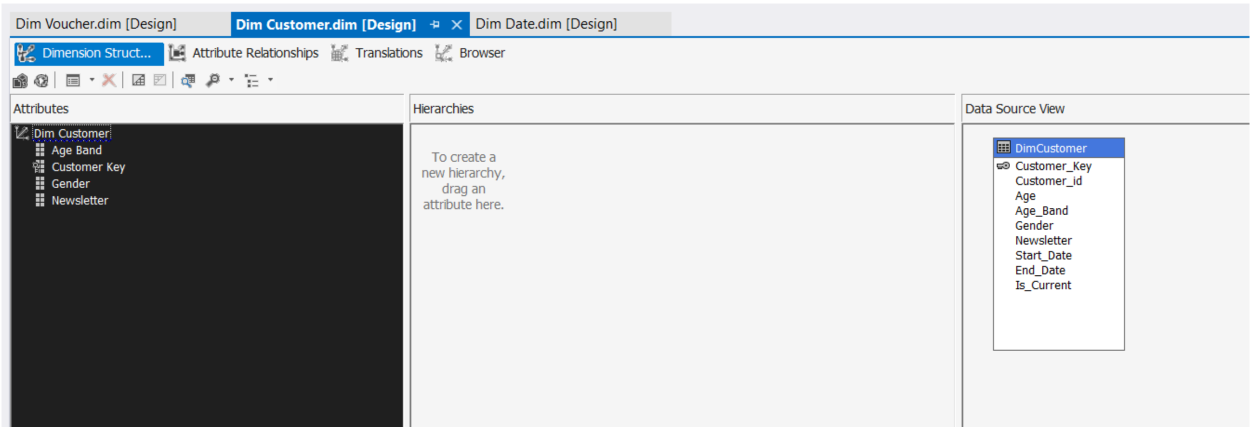
2.6.1. Creation of the Date Hierarchy

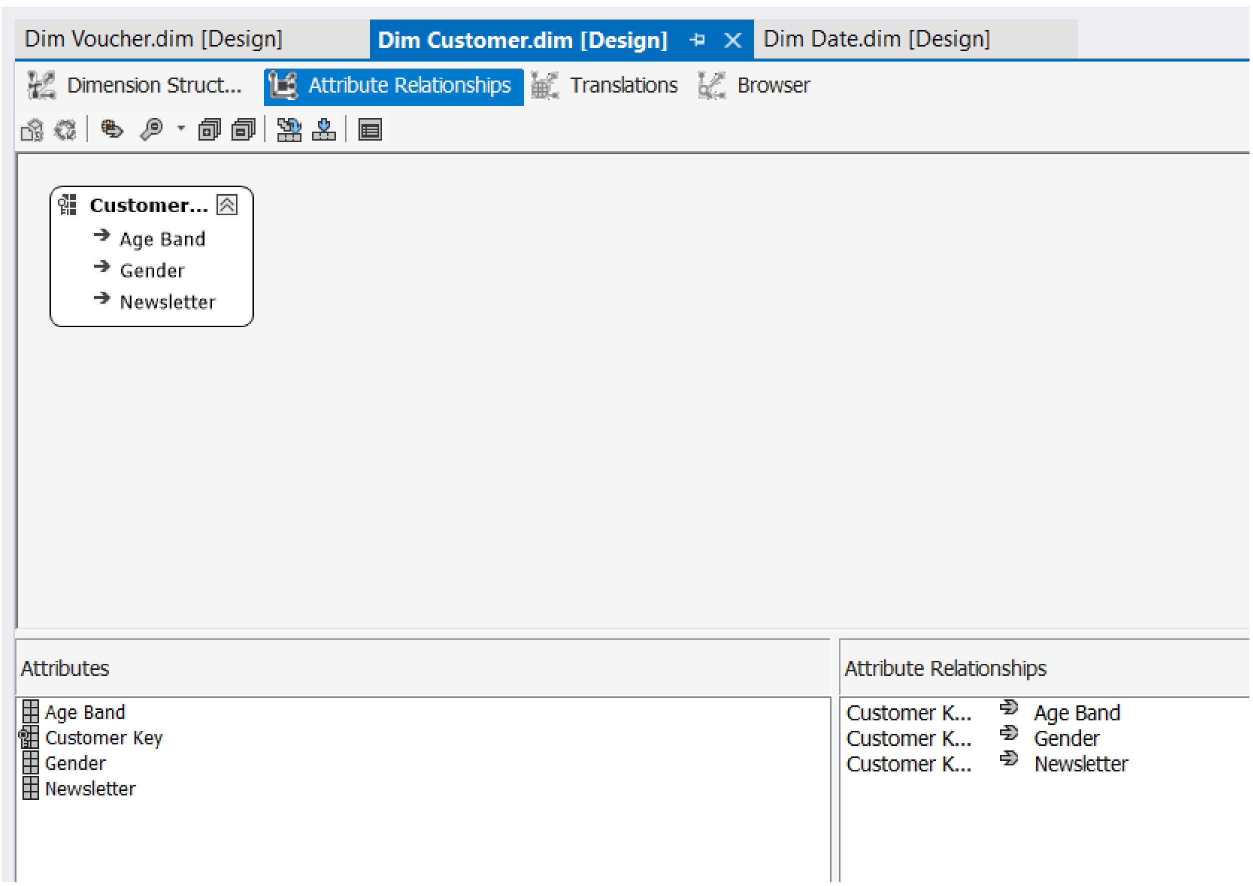




- These images show the creation of the first hierarchy in the **Dim Date** dimension of the SSAS project. A **Calendar** hierarchy was built using **Year No, Quarter No, Month Name,** and **Full Date**, and the attribute relationships were then arranged to support proper drill-down and efficient time-based analysis.

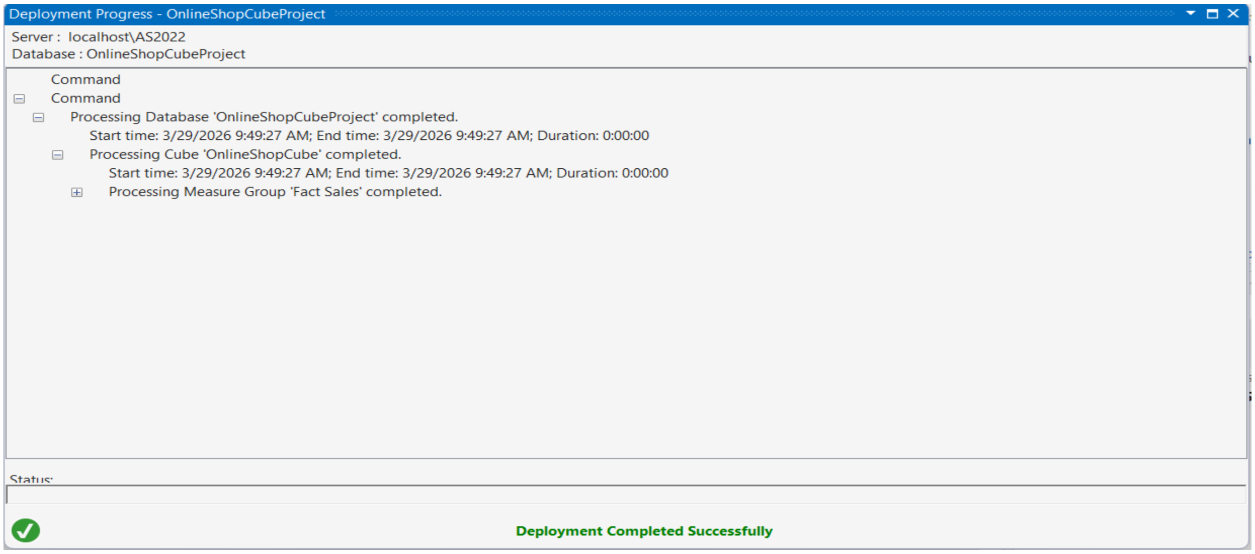
2.6.2. Creation of the Customer Hierarchy





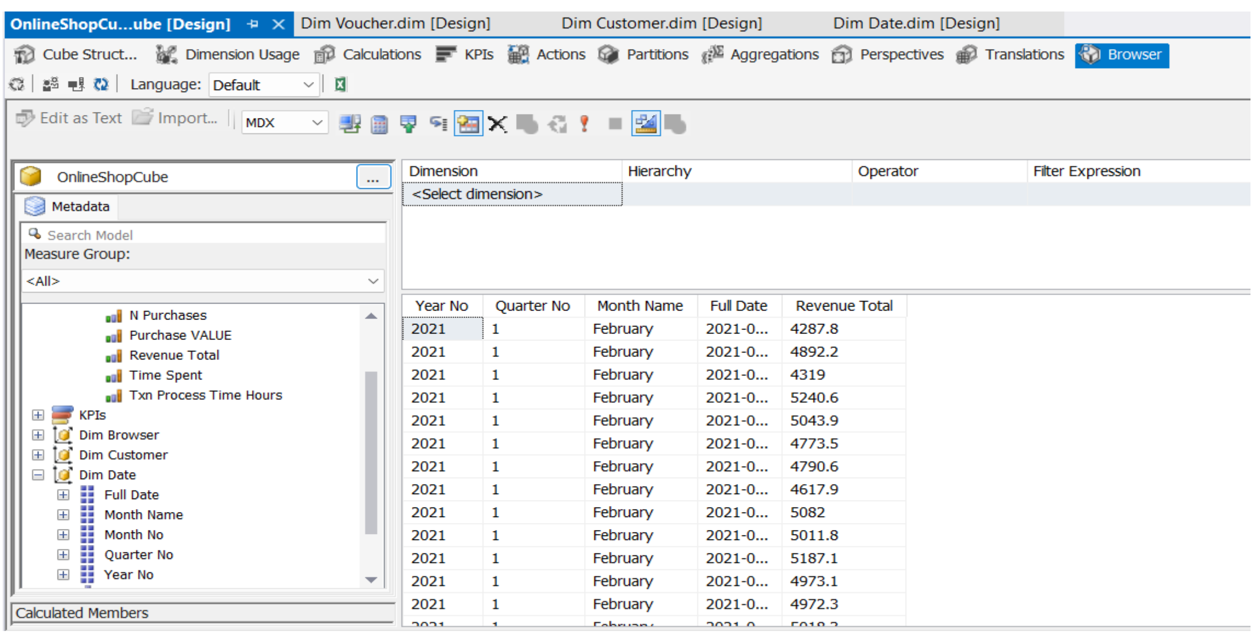
- These images show the design of the **Dim Customer** dimension for the second hierarchy in the SSAS project. The customer-related attributes such as **Age Band, Gender,** and **Newsletter** were selected, and their attribute relationships were configured to support organized customer-based analysis in the cube.

2.7.Deploying the project

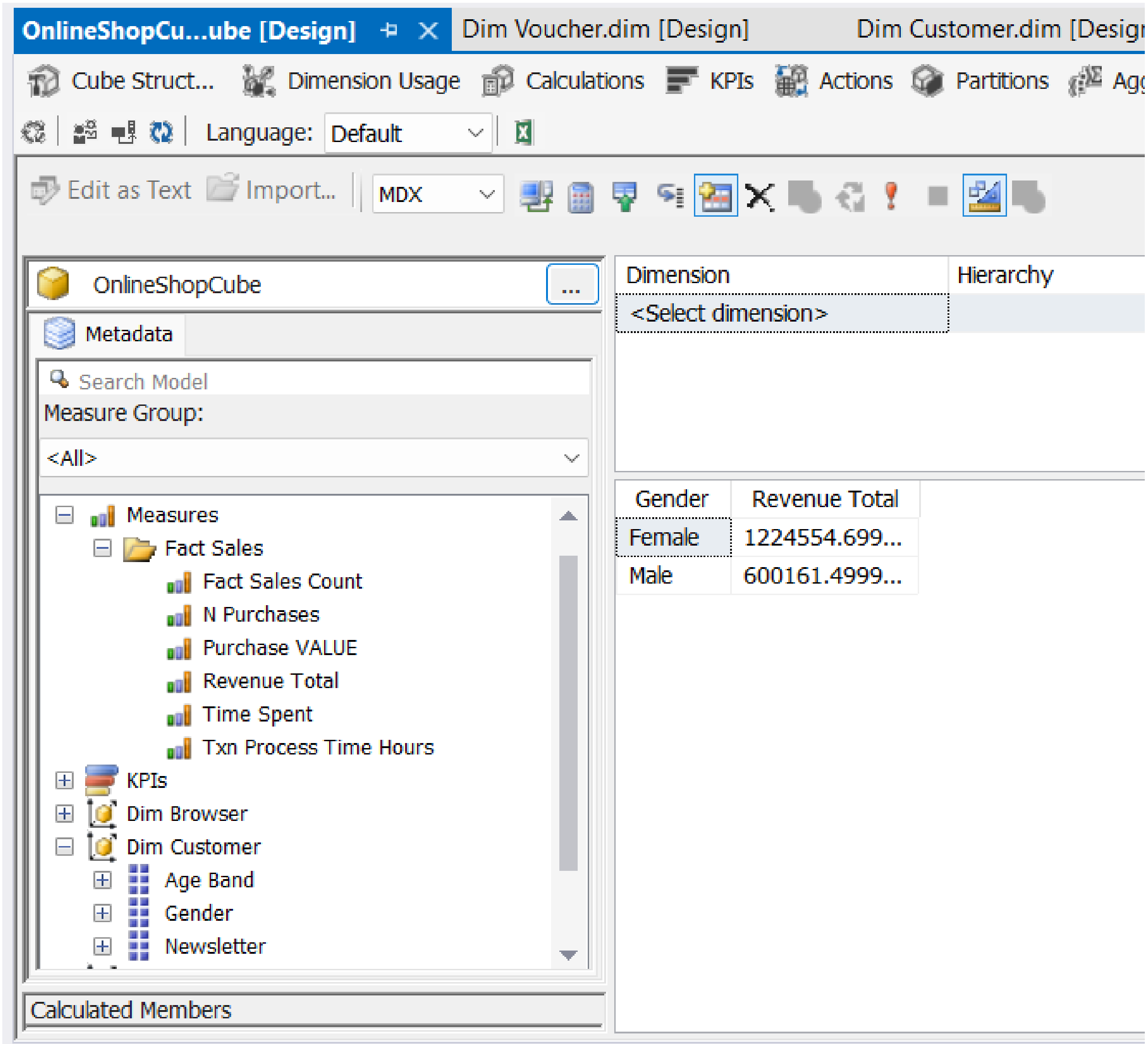


- This screen shows the **deployment progress** of the SSAS cube project. It confirms that the **OnlineShopCubeProject**, cube, and **Fact Sales** measure group were processed successfully, indicating that the cube was deployed without errors.

2.8.Process and browse the cube



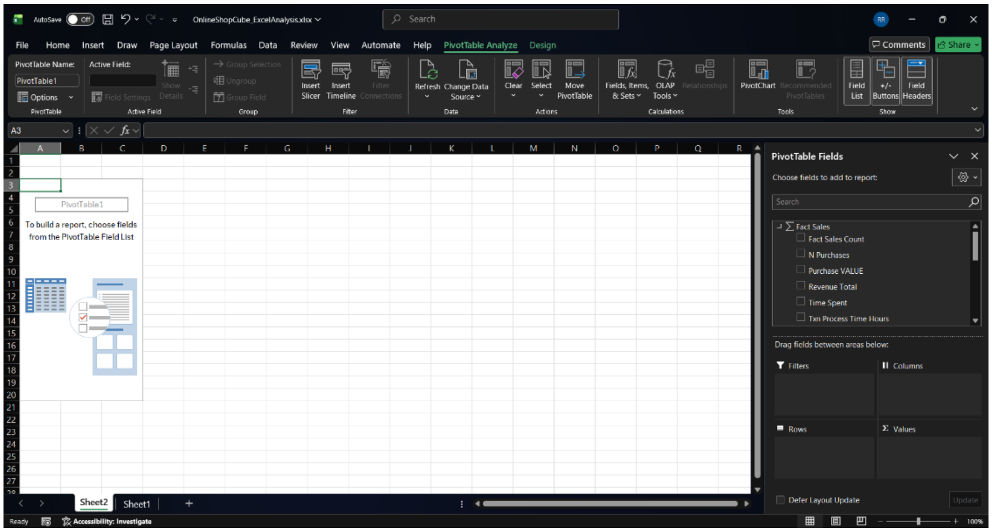
- Browser view of the SSAS cube showing successful query execution with Revenue Total across date-related attributes.



- Browser view of the SSAS cube showing Revenue Total analyzed by customer attribute Gender.

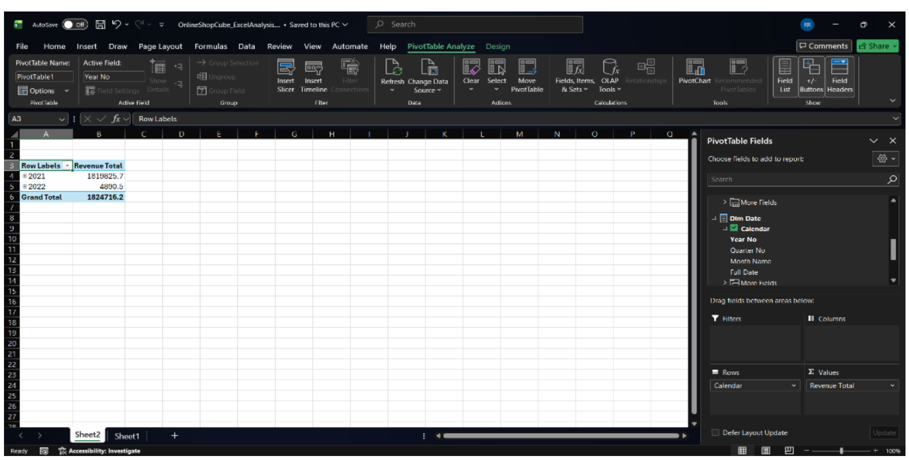
3. Demonstration of OLAP operations in Excel.

3.1. Confirm that the PivotTable field list appears



- This screen shows the **Excel PivotTable interface** connected to the deployed SSAS cube. The available measures from **Fact Sales** are displayed in the PivotTable Fields pane, allowing interactive analysis by dragging fields into rows, columns, filters, and values.

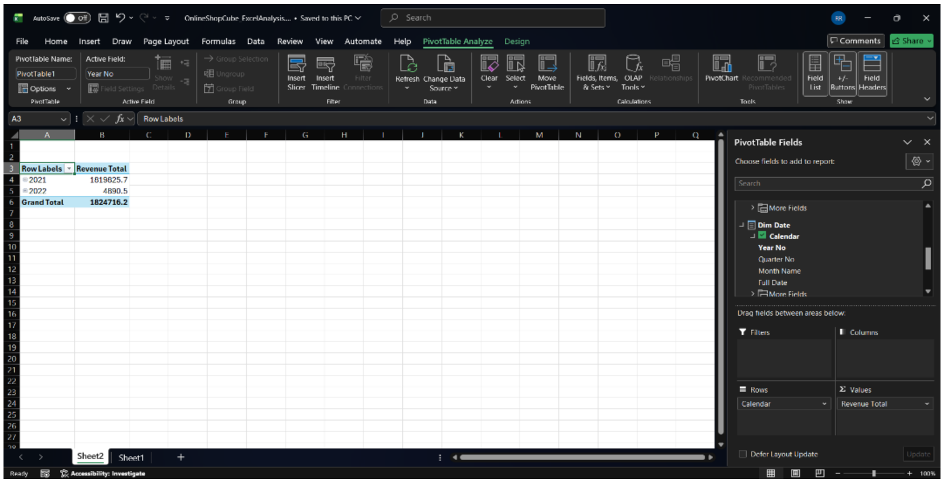
3.2. Create the base PivotTable



- This screen shows an **Excel PivotTable report** created from the SSAS cube. It analyzes **Revenue Total** by the **Calendar** hierarchy, displaying the revenue values by year and the overall grand total for quick time-based analysis.

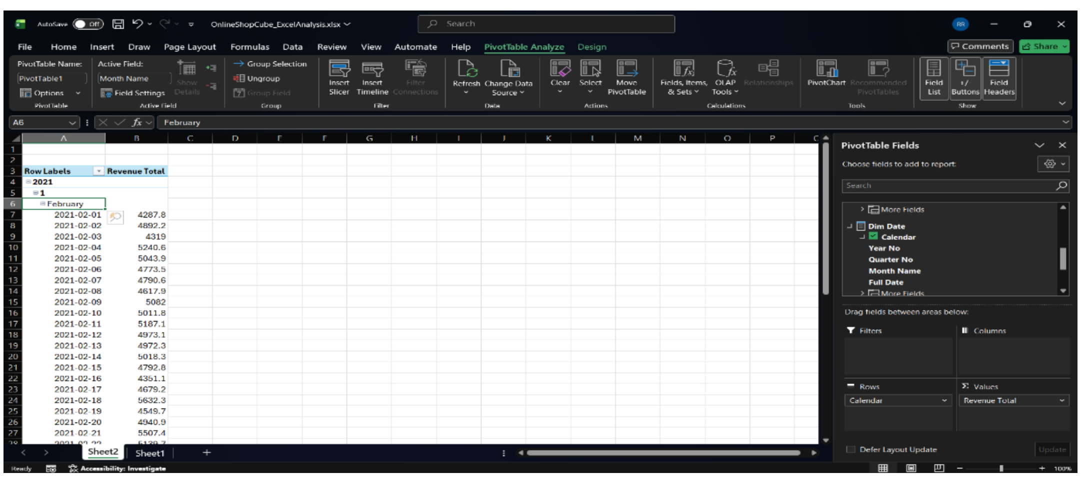
3.3.Demonstrate the 5 OLAP operations

3.3.1. Roll-up



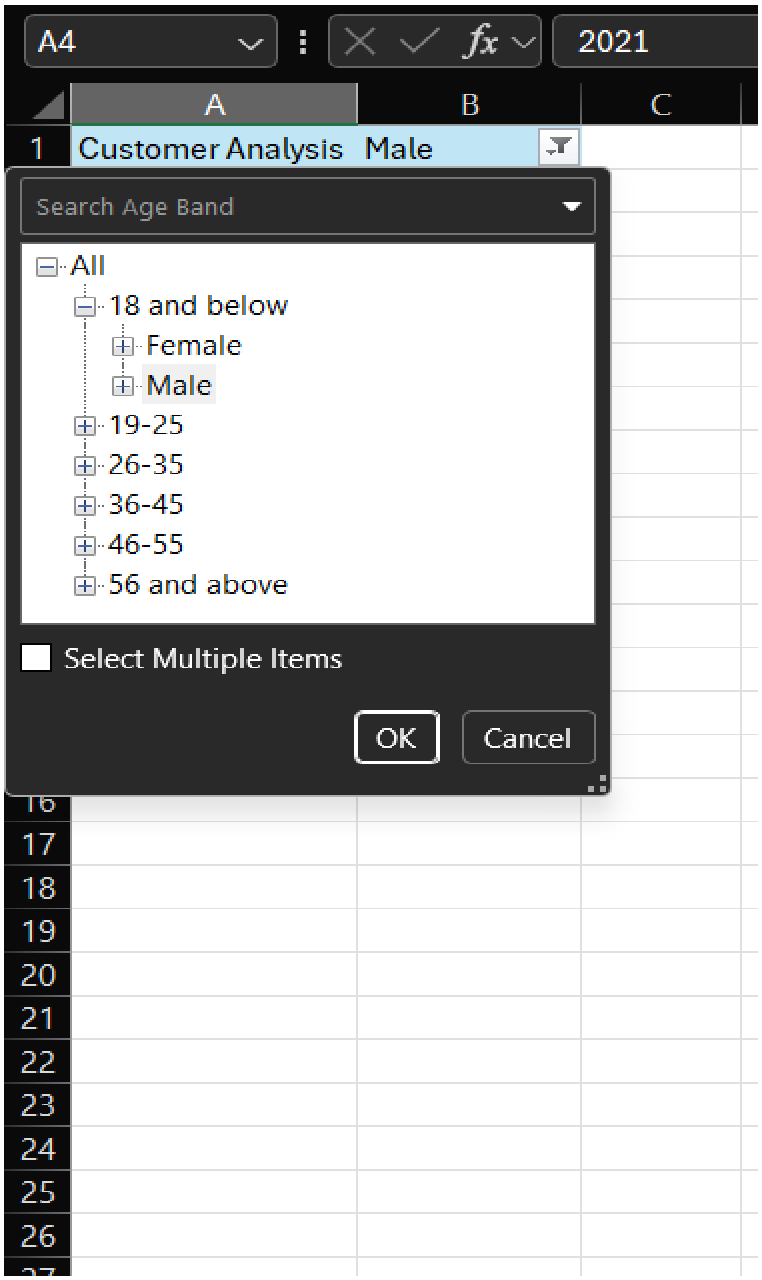
- This screen demonstrates the **roll-up OLAP operation** in the Excel PivotTable connected to the SSAS cube. The data has been aggregated to the **Year** level in the **Calendar** hierarchy, showing summarized **Revenue Total** values for each year instead of lower-level details such as months or dates.

3.3.2. Drill-down



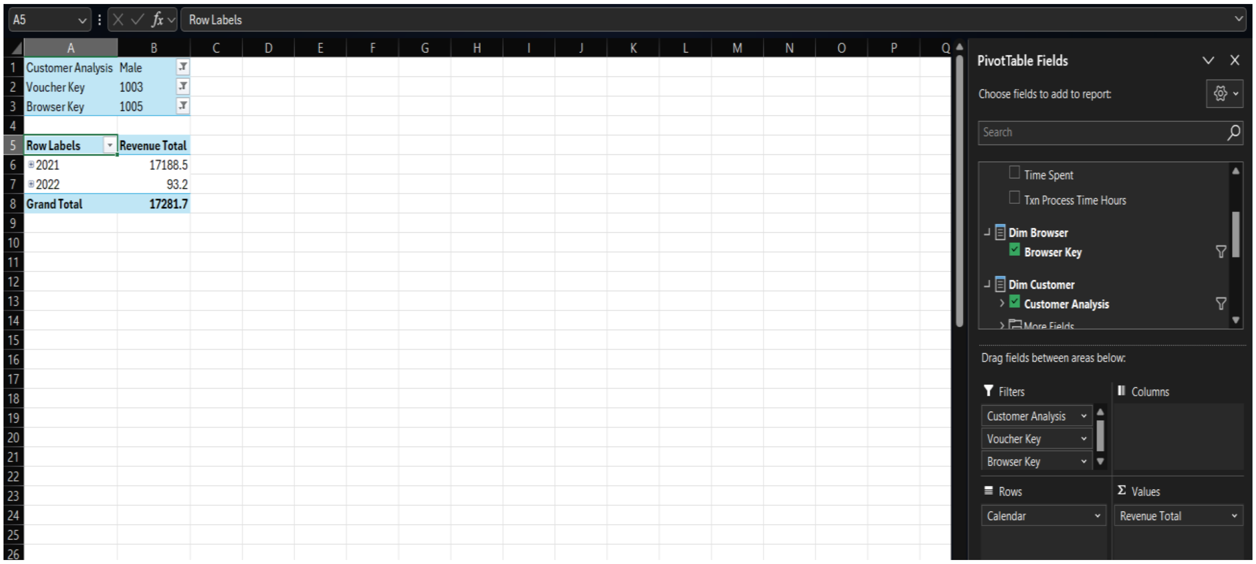
- **This screen demonstrates the** drill-down OLAP operation **in the Excel PivotTable connected to the SSAS cube. The analysis has moved from the higher** Year **level down to** Quarter, Month, **and finally** Full Date**, allowing detailed viewing of** Revenue Total **for individual days within February 2021.**

3.3.3. Slice



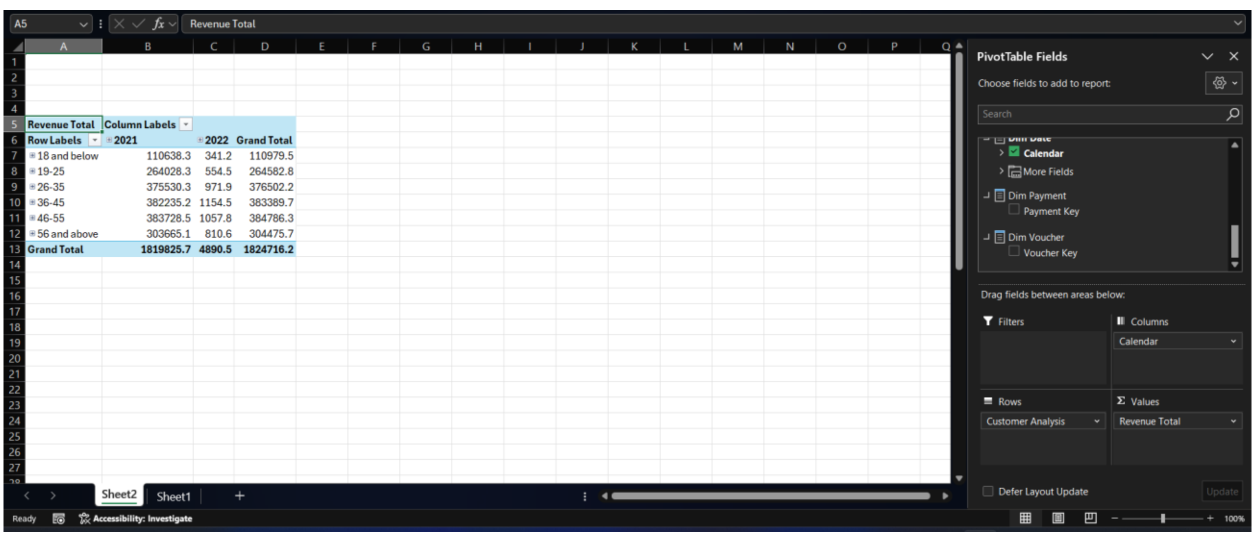
- Slice operation showing cube data filtered to a single dimension value.

3.3.4. Dice



- Dice operation showing data filtered using multiple dimensions simultaneously.

3.3.5. Pivot

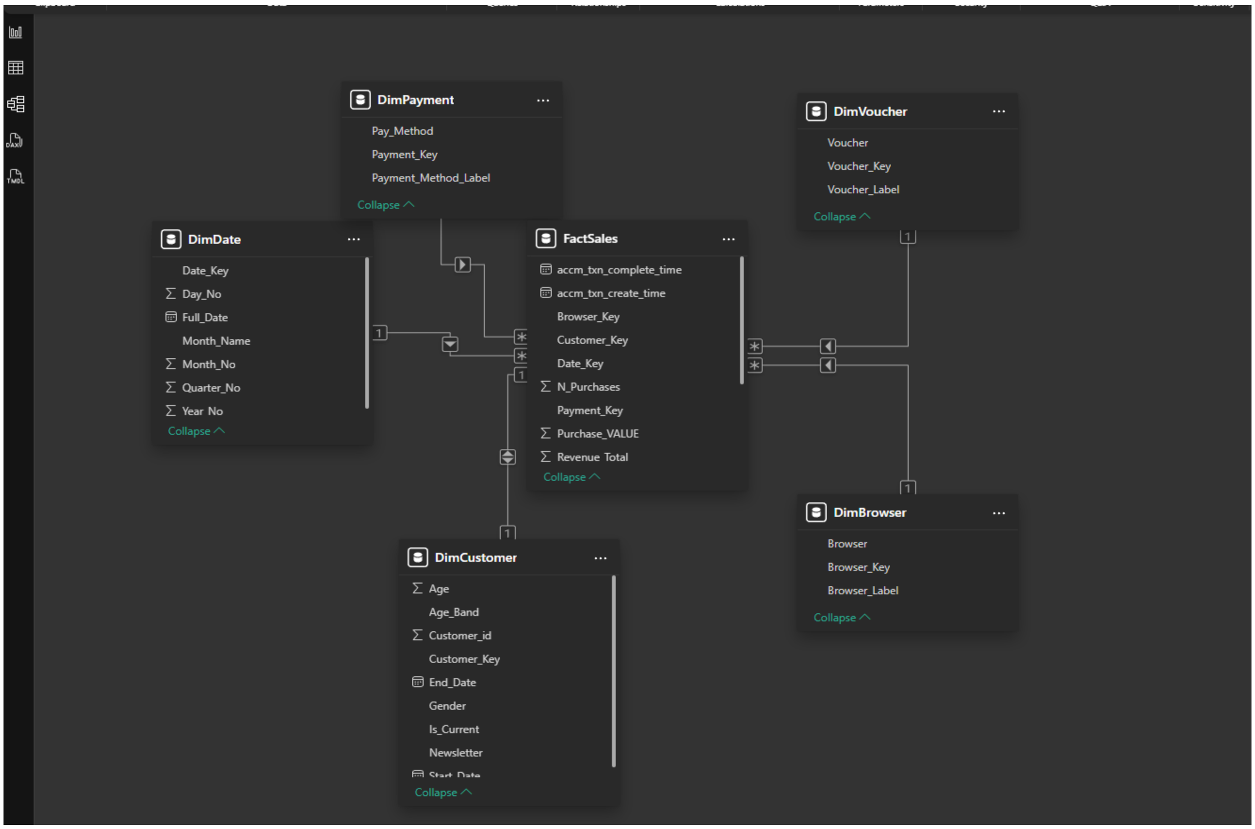


Revenue Total	Column Labels		
Row Labels	2021	2022	Grand Total
18 and below	110638.3	341.2	110979.5
19-25	264028.3	554.5	264582.8
26-35	375530.3	971.9	376502.2
36-45	382235.2	1154.5	383389.7
46-55	383728.5	1057.8	384786.3
56 and above	303665.1	810.6	304475.7
Grand Total	1819825.7	4890.5	1824716.2

- Pivot operation showing the same measure analyzed by changing row and column orientation.

4. PowerBI Reports

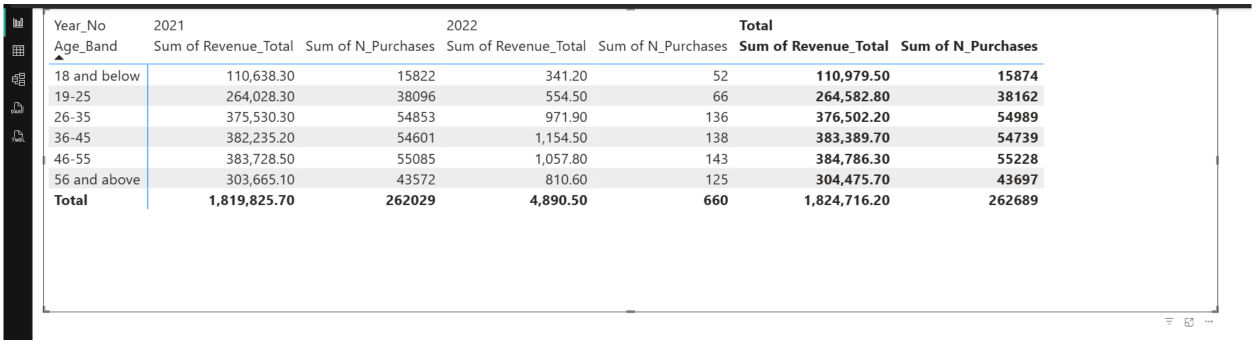
4.1. **Data Model View** of the OnlineShop analytical dataset.



- This screen shows the **data model view** of the OnlineShop analytical dataset. It presents the **FactSales** table at the center connected to the related dimension tables such as **DimDate, DimCustomer, DimPayment, DimVoucher,** and **DimBrowser**, forming a star schema for analysis.

4.2. Develop reports in PowerBI

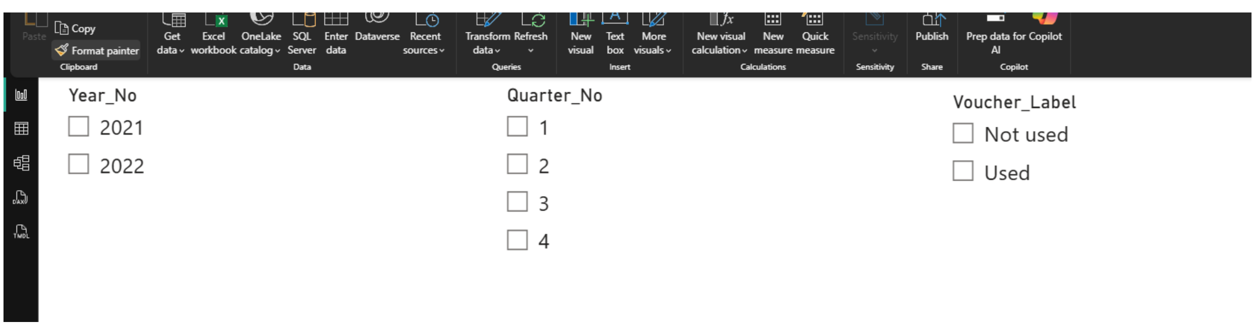
4.2.1. Report 1 – Matrix



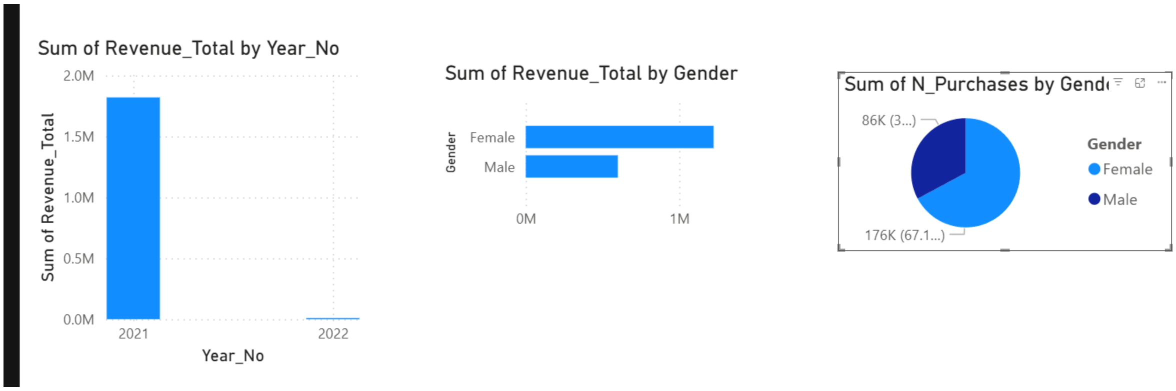
Year_No	2021		2022		Total	
Age_Band	Sum of Revenue_Total	Sum of N_Purchases	Sum of Revenue_Total	Sum of N_Purchases	**Sum of Revenue_Total**	**Sum of N_Purchases**
18 and below	110,638.30	15822	341.20	52	**110,979.50**	**15874**
19-25	264,028.30	38096	554.50	66	**264,582.80**	**38162**
26-35	375,530.30	54853	971.90	136	**376,502.20**	**54989**
36-45	382,235.20	54601	1,154.50	138	**383,389.70**	**54739**
46-55	383,728.50	55085	1,057.80	143	**384,786.30**	**55228**
56 and above	303,665.10	43572	810.60	125	**304,475.70**	**43697**
Total	**1,819,825.70**	**262029**	**4,890.50**	**660**	**1,824,716.20**	**262689**

- This image shows the first **Power BI matrix report** created from the OnlineShop data model. It presents **Revenue Total** and **Number of Purchases** by **Age Band** across **2021, 2022,** and the overall **Total**, making it easy to compare customer segment performance over time.

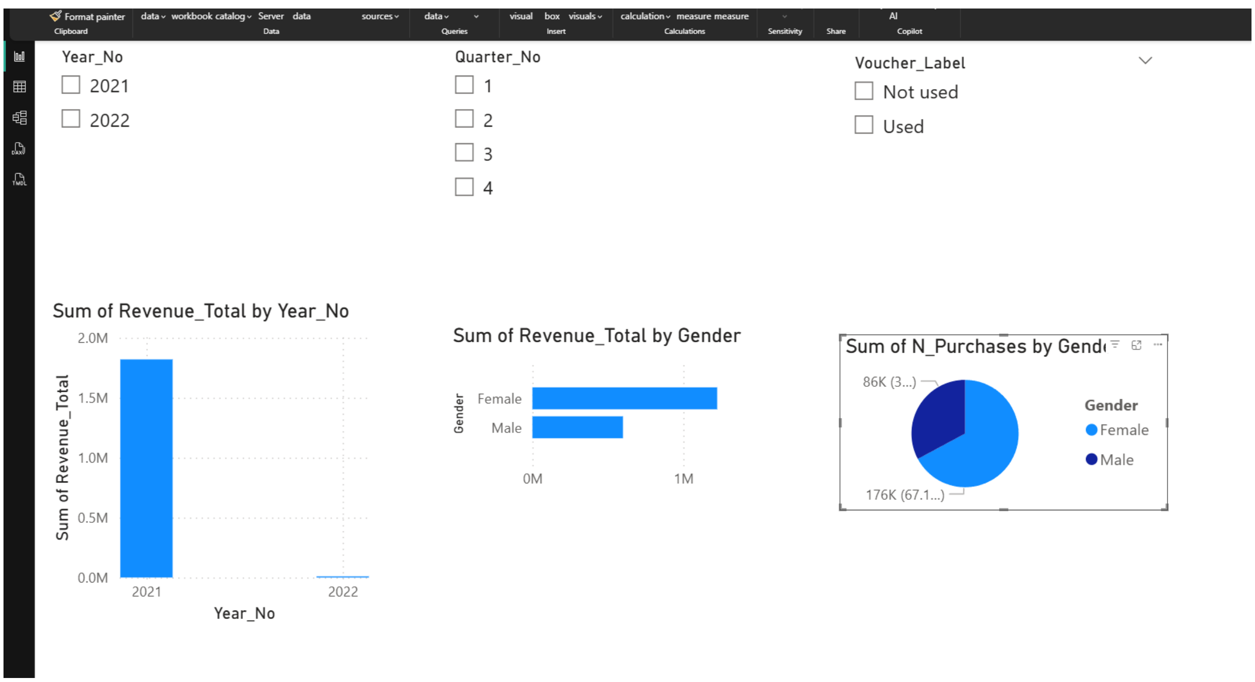
4.2.2. Report 2 – Slicers

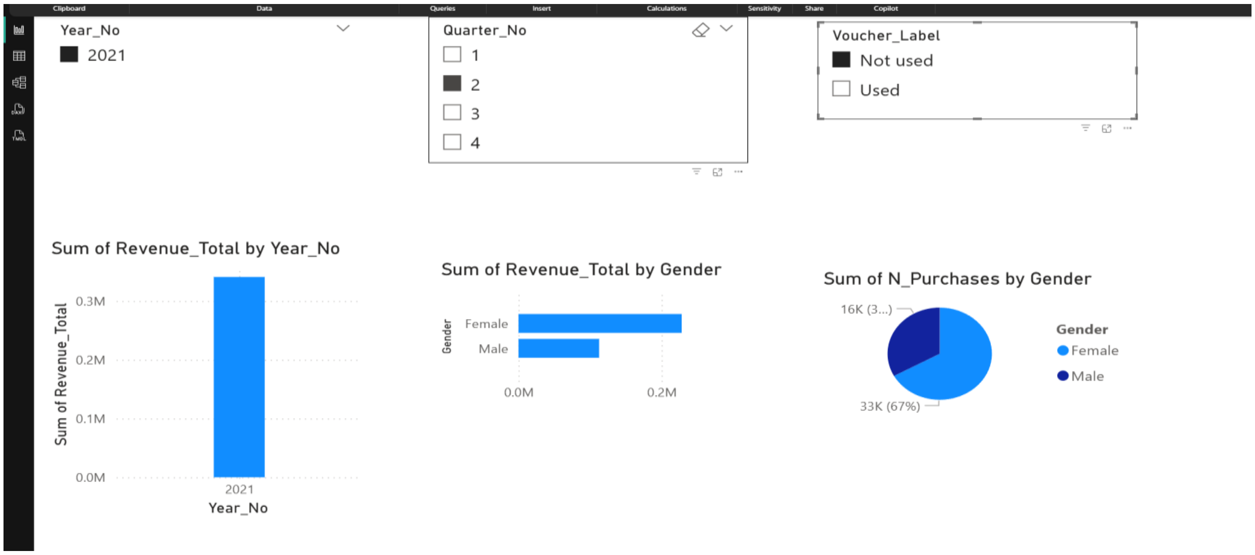


- This image shows the **Power BI slicers** created for the second report. The slicers allow users to interactively filter the analysis by **Year**, **Quarter**, and **Voucher Label**, making the report more flexible and user-friendly.

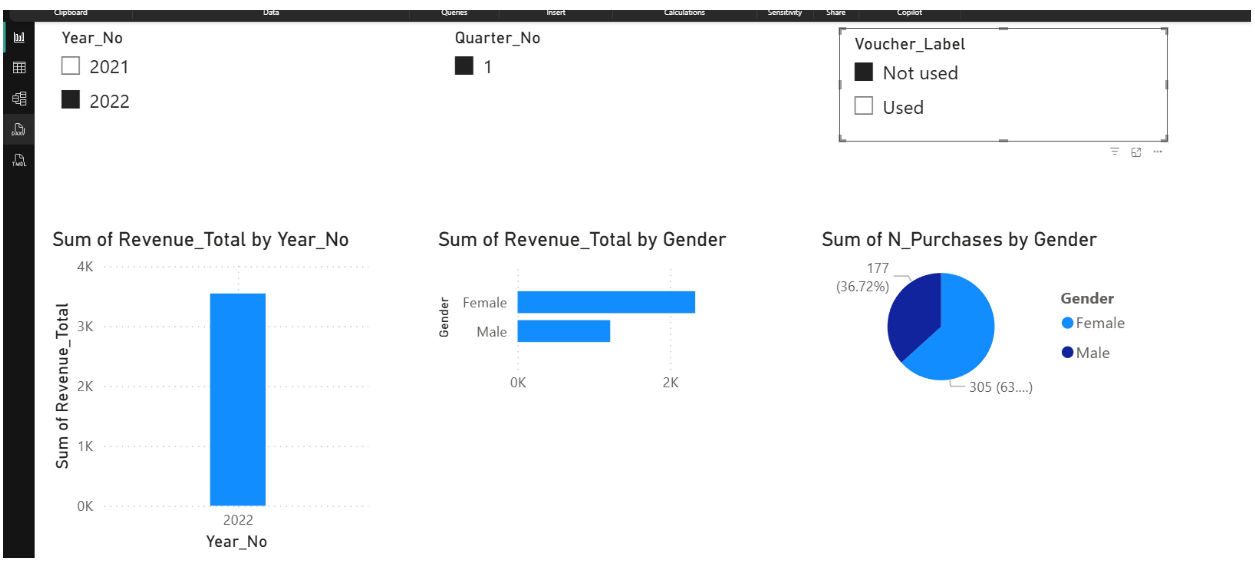


- This image shows the **visuals used in Report 2** to demonstrate that the slicers are working correctly. The charts display **Revenue Total by Year**, **Revenue Total by Gender**, and **Number of Purchases by Gender**, updating dynamically based on the slicer selections.



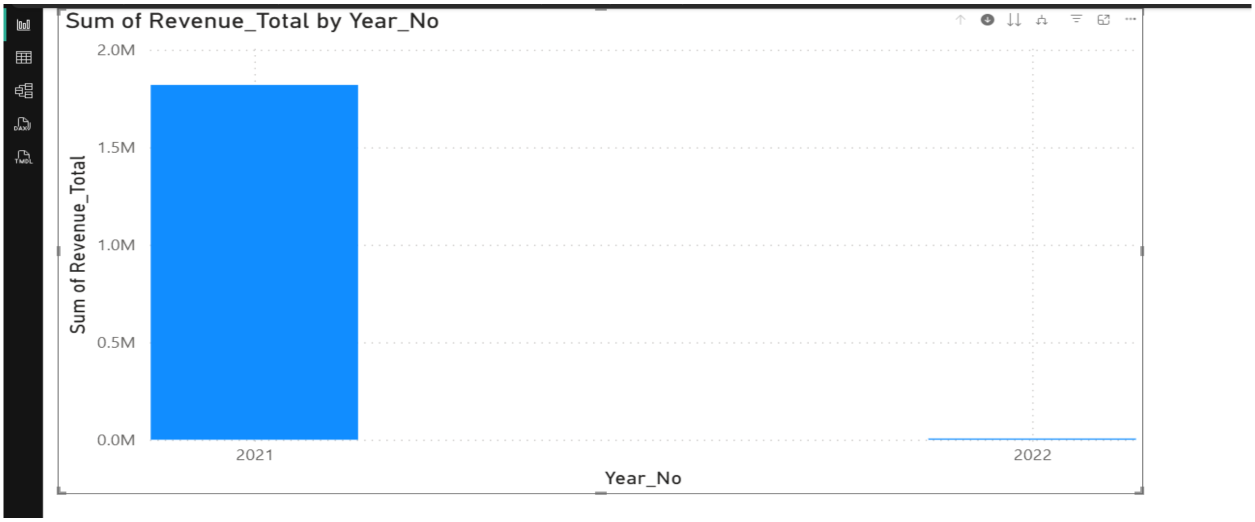


- This screen shows **Report 2 after applying slicer selections** in Power BI. The visuals have been updated to display results only for **Year 2021**, **Quarter 2**, and **Voucher Label: Not used**, confirming that the slicers are filtering the report correctly.

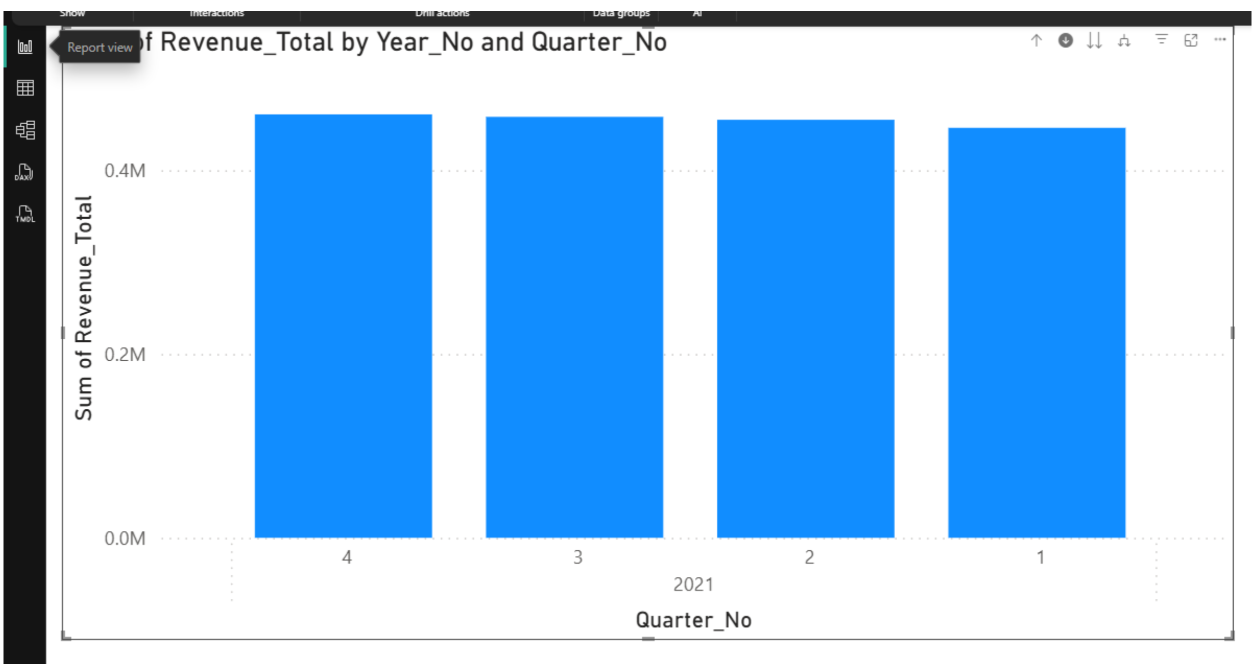


- This screen shows another filtered view of **Report 2** in Power BI after applying different slicer selections. The visuals now display results for **Year 2022**, **Quarter 1**, and **Voucher Label: Not used**, illustrating how the slicers dynamically change the analysis output.

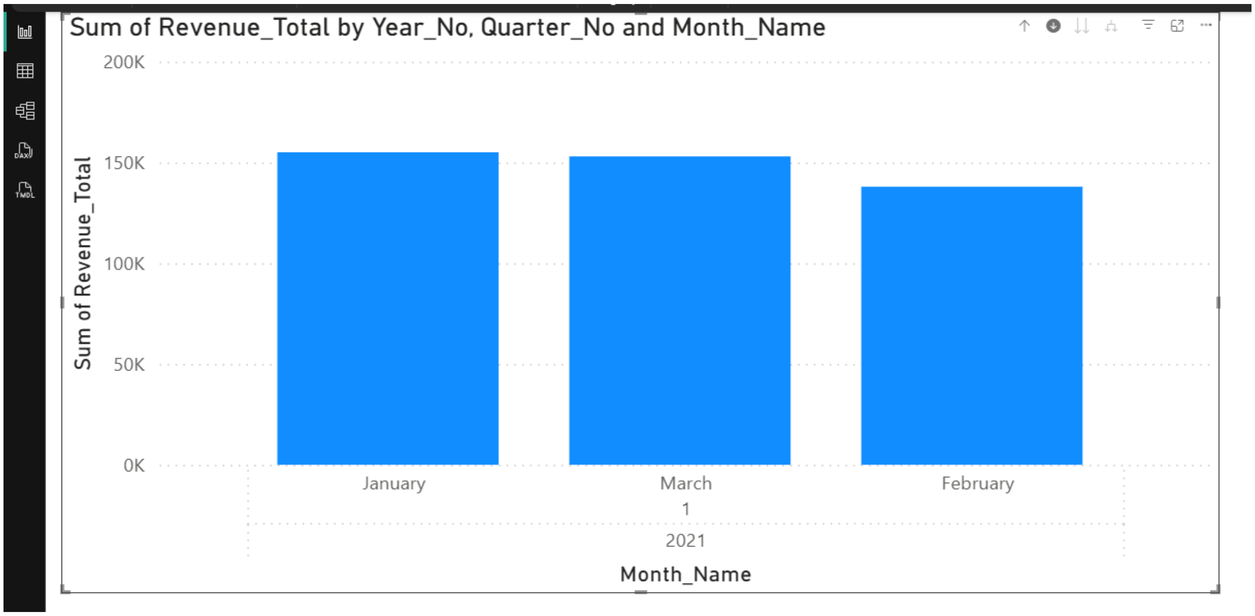
4.2.3. Report 3 - Drill Down



- This visual shows a **Power BI column chart** analyzing **Revenue Total by Year**. It highlights that revenue is much higher in **2021** compared to **2022**, making yearly performance comparison clear and easy to understand.

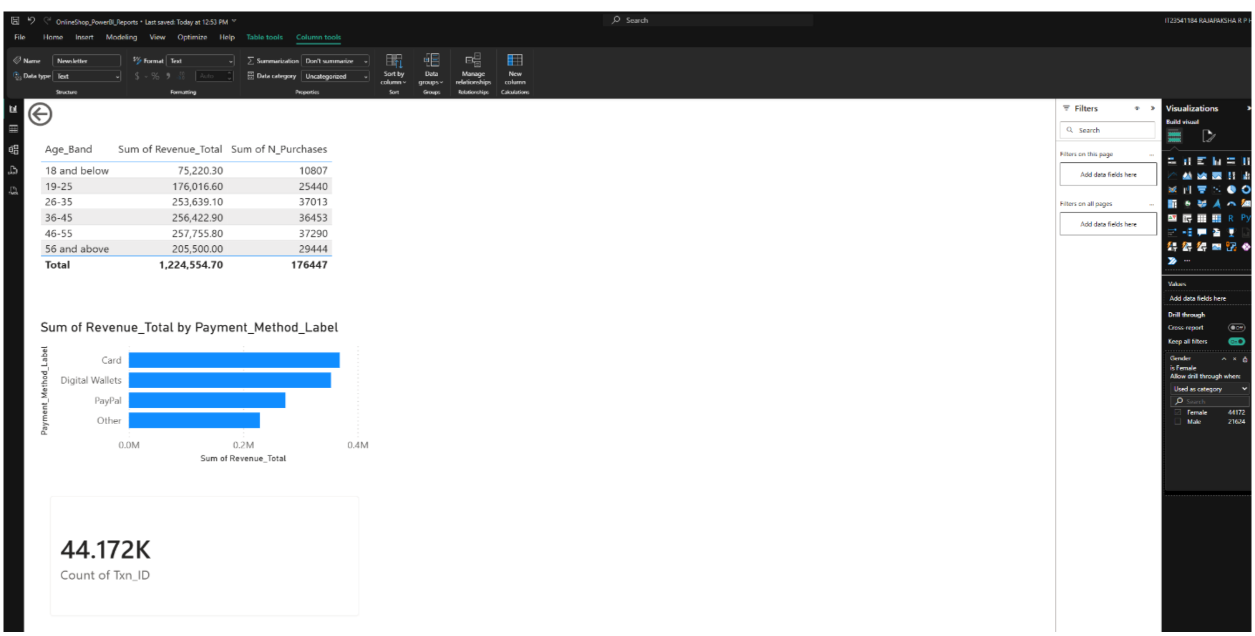


- This visual shows a **Power BI chart of Revenue Total by Year and Quarter**. It breaks down the revenue for **2021** across the four quarters, allowing quarter-wise comparison to observe how revenue changed within the year.



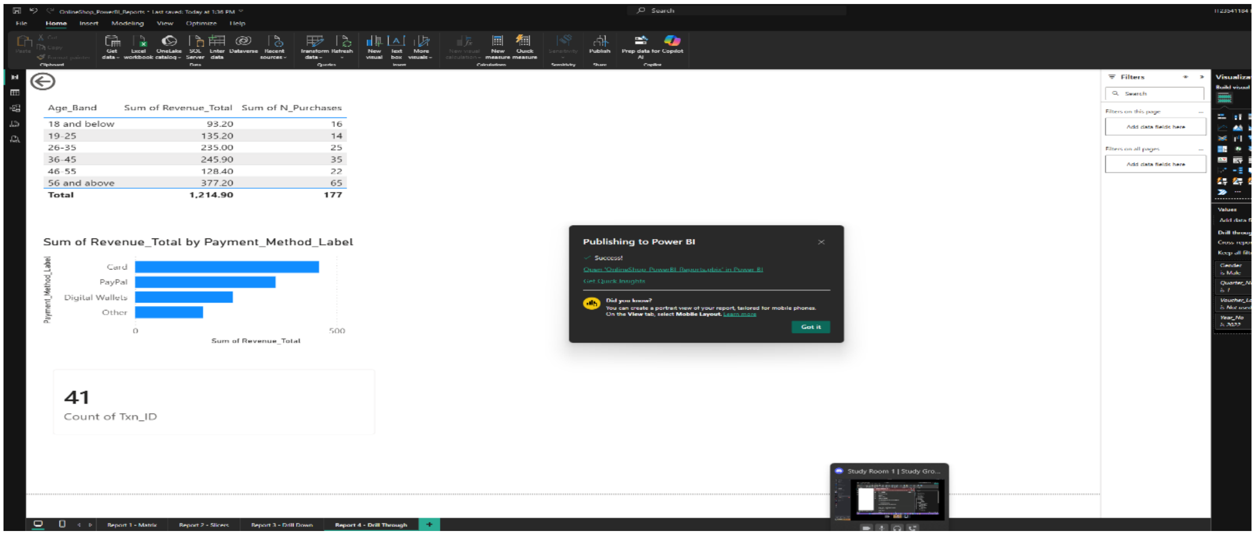
- This visual shows a **Power BI chart of Revenue Total by Year, Quarter, and Month**. It drills the analysis further down to the **monthly level** for **Quarter 1 of 2021**, enabling a more detailed comparison of revenue across January, February, and March.

4.2.4. Report 4 - Drill Through

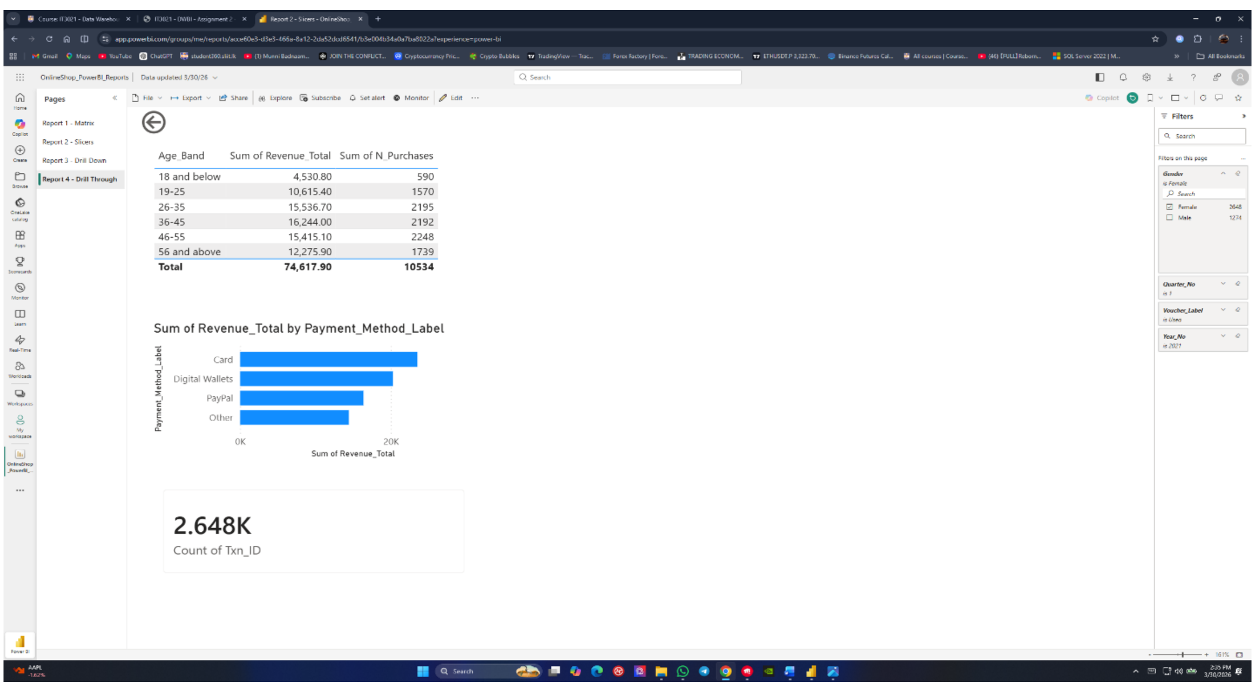


- This screen shows **Report 4 using drill-through** in Power BI after receiving the filter context from the slicers. It presents a detailed view for the selected conditions, with filtered summaries by **Age Band**, **Payment Method**, and **transaction count**, allowing deeper analysis of the chosen subset of data.

4.3. Publishing the Reports Developed



- This screen shows the **successful publishing of the Power BI report** to the Power BI Service. It confirms that the completed report file was uploaded successfully, making the dashboards and drill-through analysis available for online access and sharing.



- This screen shows the **online Power BI report page** after the report was published successfully to the Power BI Service. It confirms that the drill-through report is now available on the web, with interactive visuals and filter options accessible for online viewing and analysis.